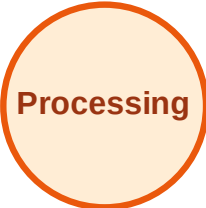


# BFS Traversal

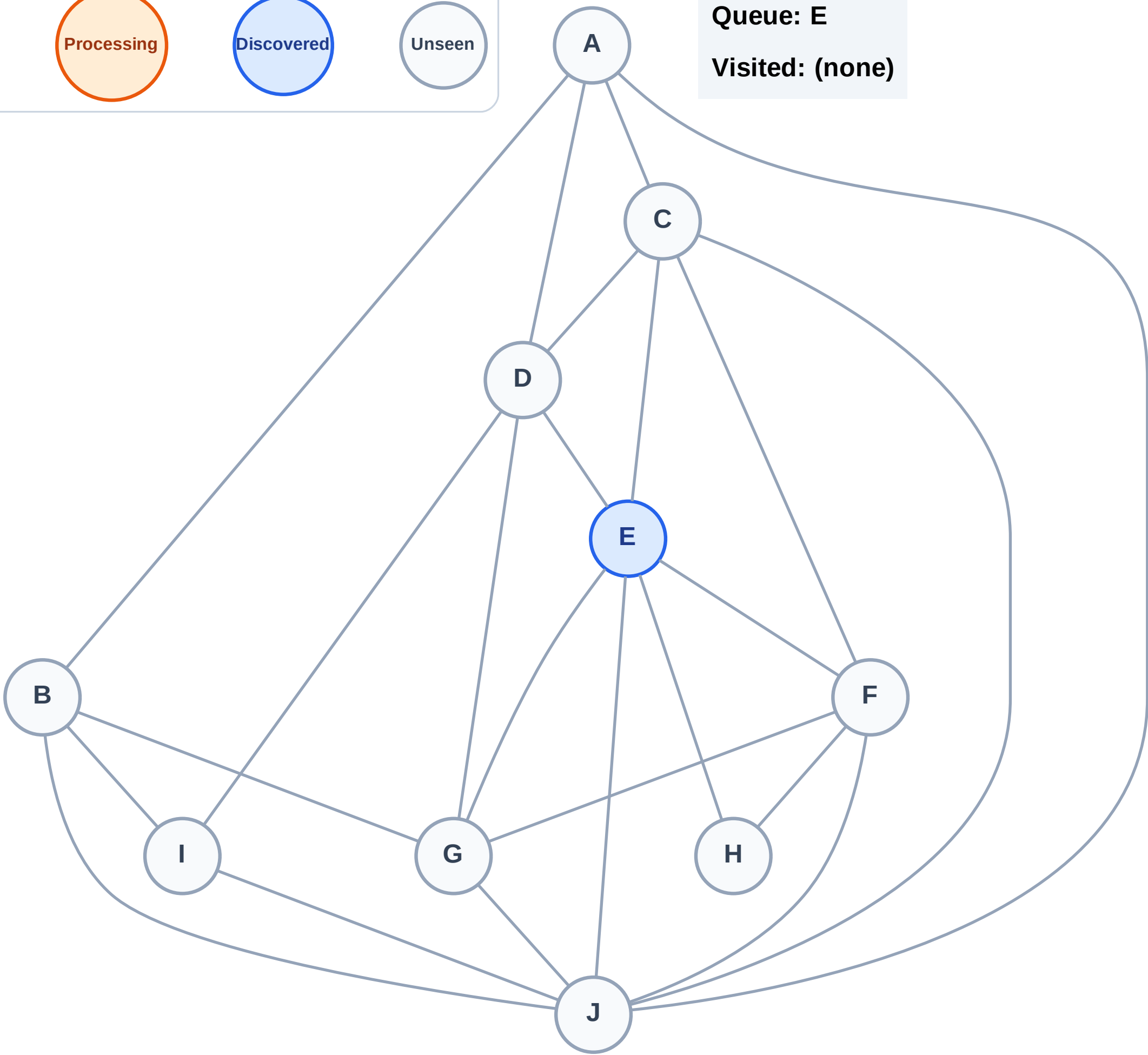
Step 1: Start at E

## Legend



Queue: E

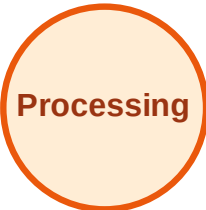
Visited: (none)



# BFS Traversal

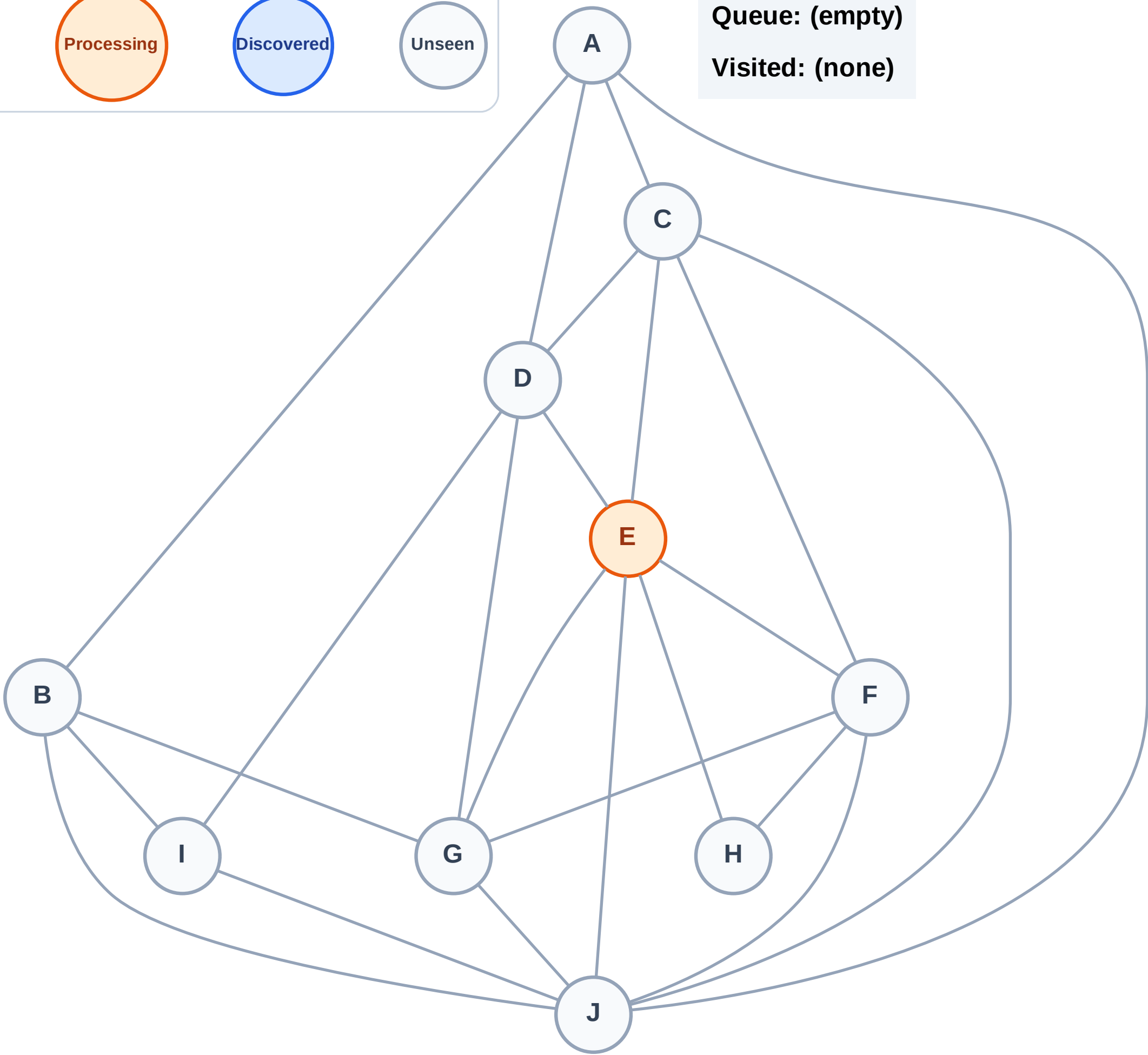
Step 2: Process node E

## Legend



Queue: (empty)

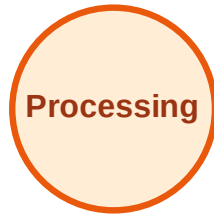
Visited: (none)



# BFS Traversal

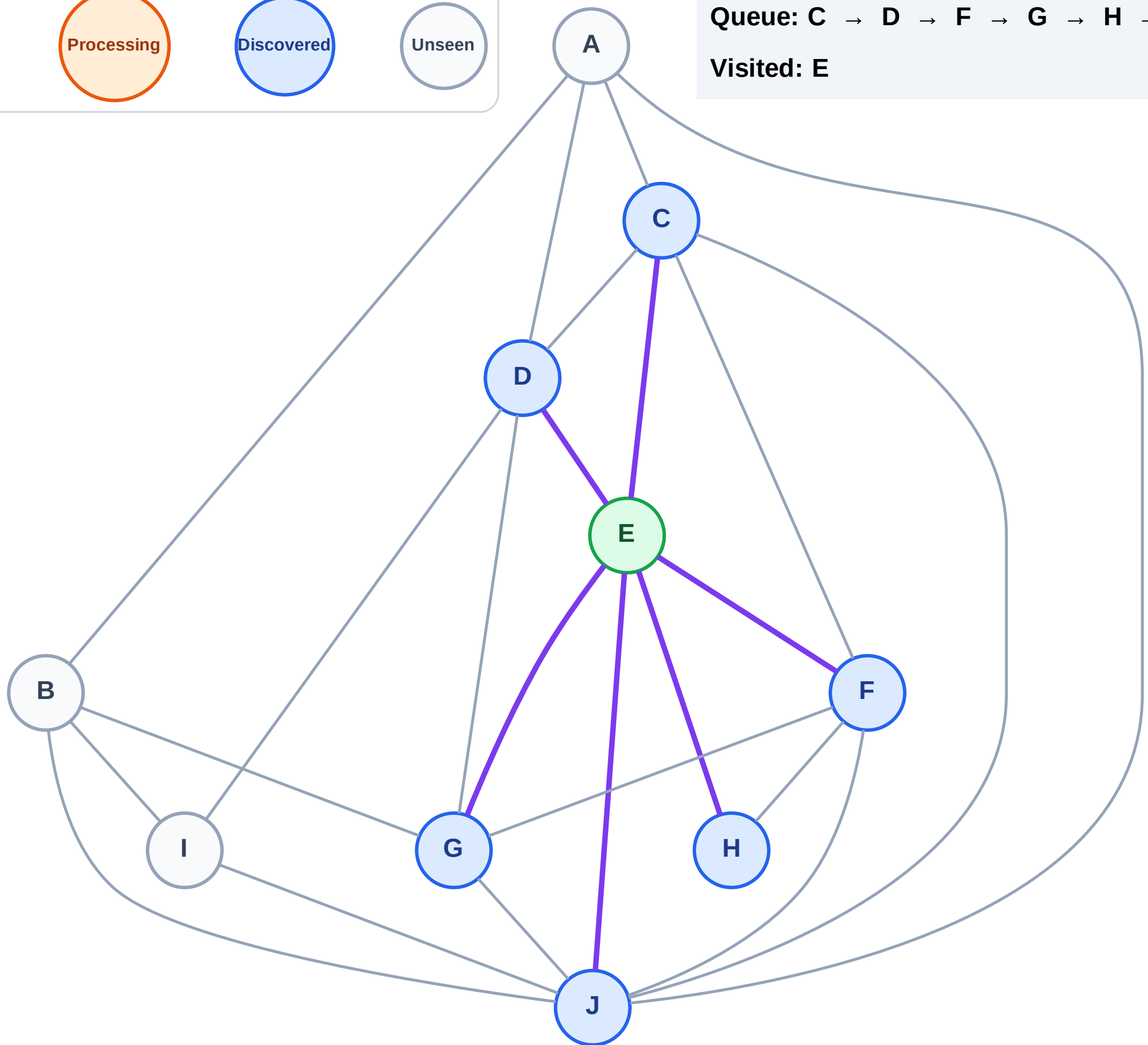
Step 3: Discover C, D, F, G, H, J from E

## Legend



Queue: C → D → F → G → H → J

Visited: E



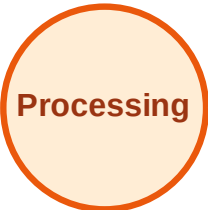
# BFS Traversal

Step 4: Process node C

## Legend



Complete



Processing



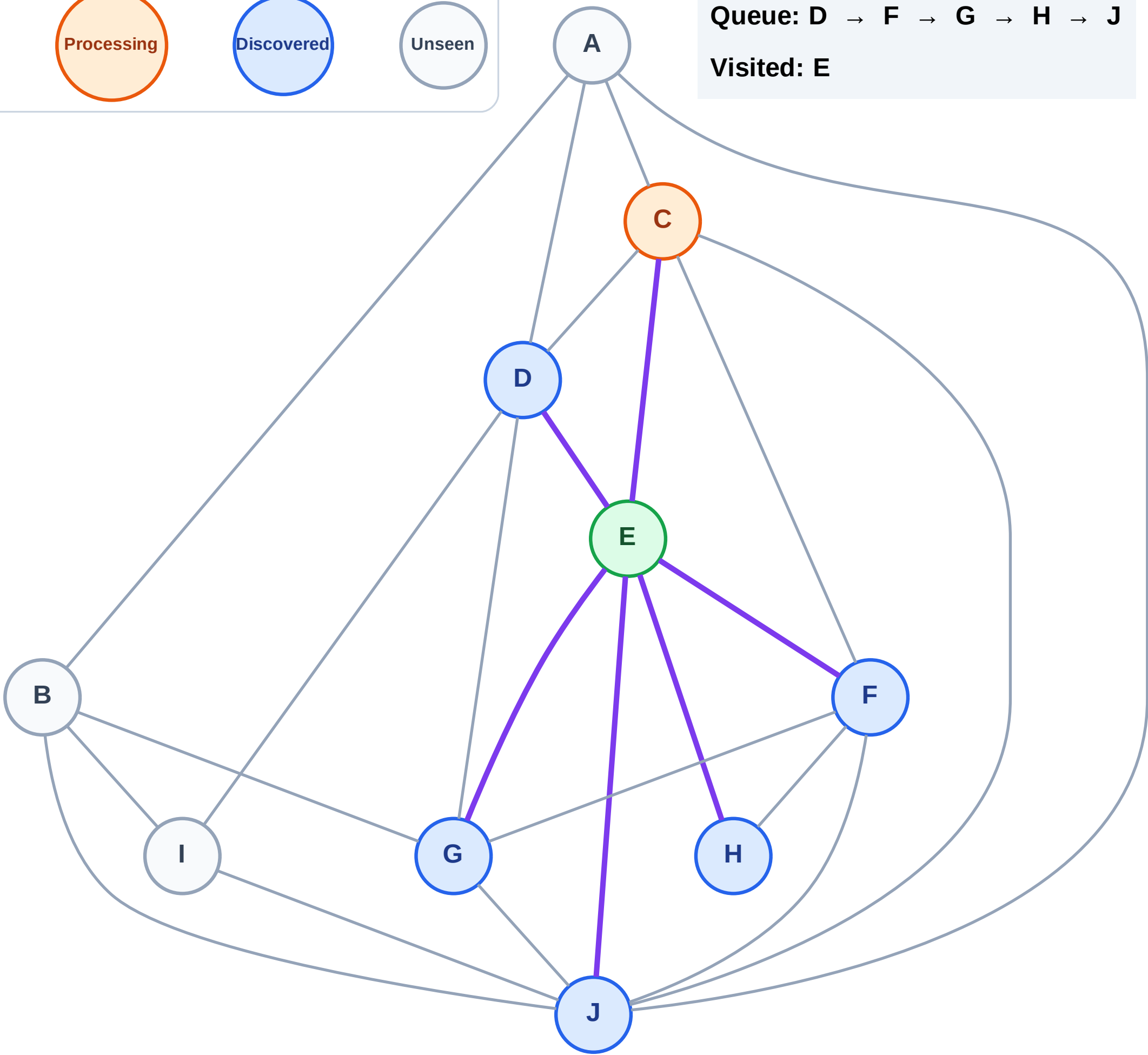
Discovered



Unseen

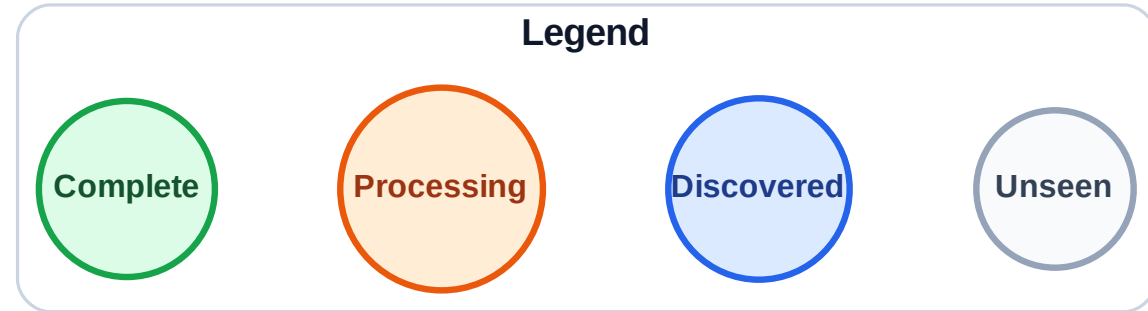
Queue: D → F → G → H → J

Visited: E



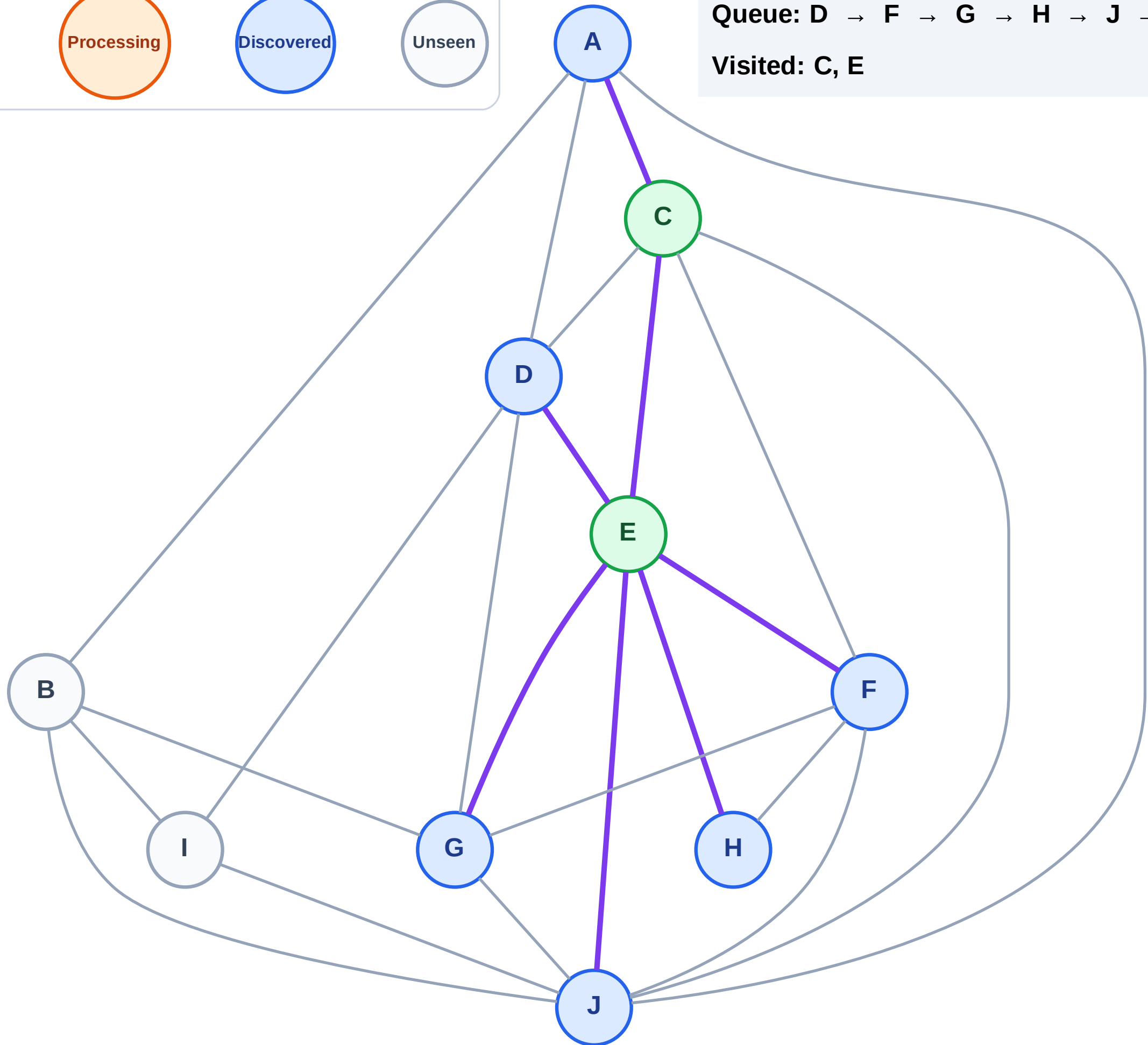
# BFS Traversal

Step 5: Discover A from C



Queue: D → F → G → H → J → A

Visited: C, E



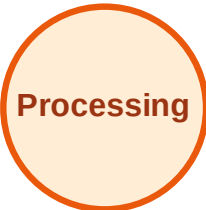
# BFS Traversal

Step 6: Process node D

## Legend



Complete



Processing



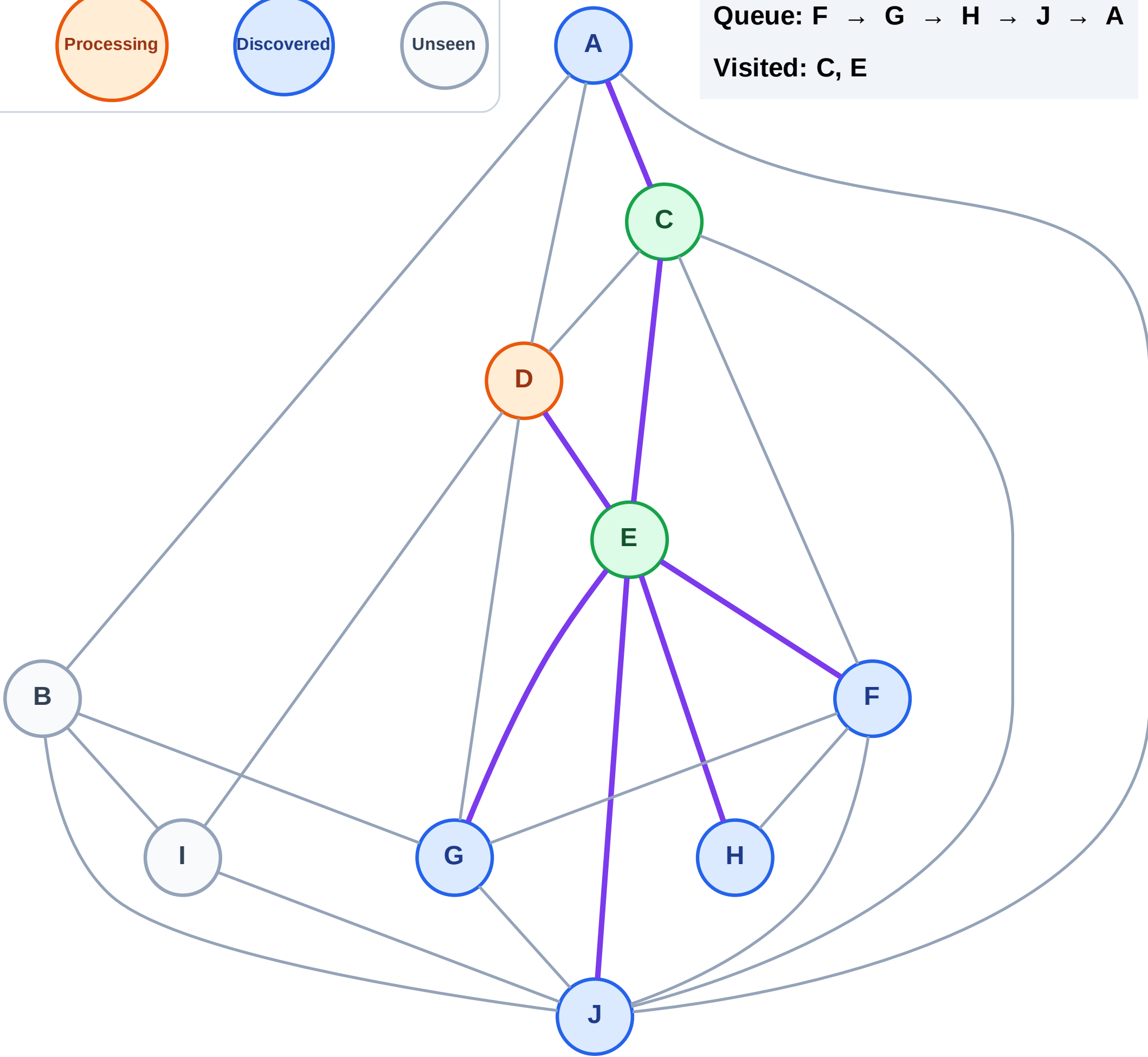
Discovered



Unseen

Queue: F → G → H → J → A

Visited: C, E



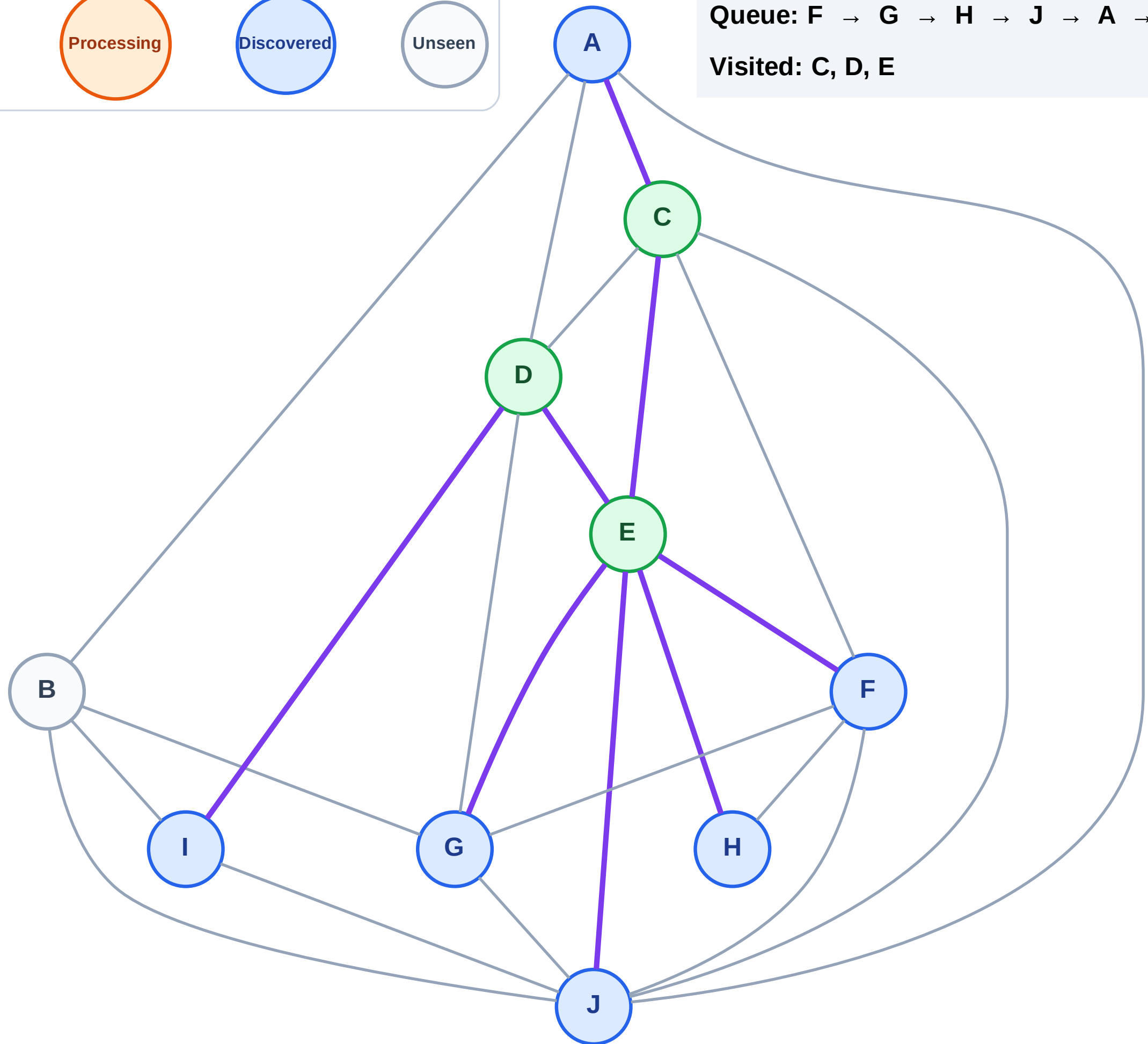
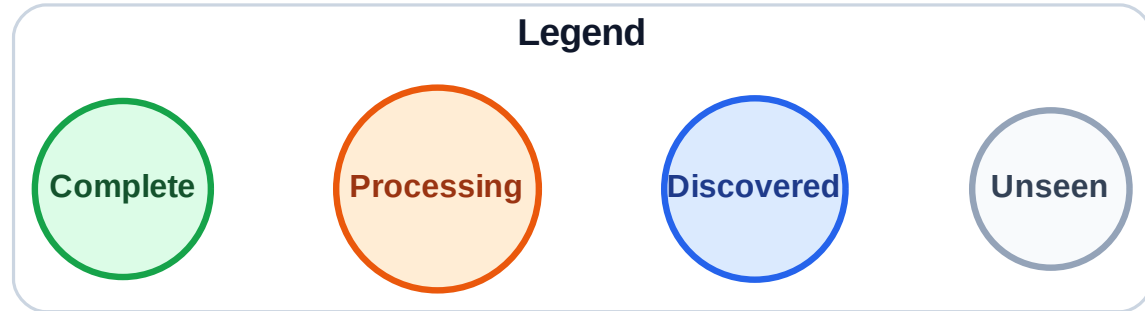


# BFS Traversal

Step 7: Discover I from D

Queue: F → G → H → J → A → I

Visited: C, D, E



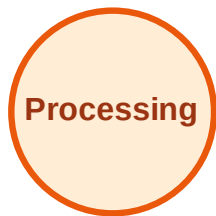
# BFS Traversal

Step 8: Process node F

## Legend



Complete



Processing



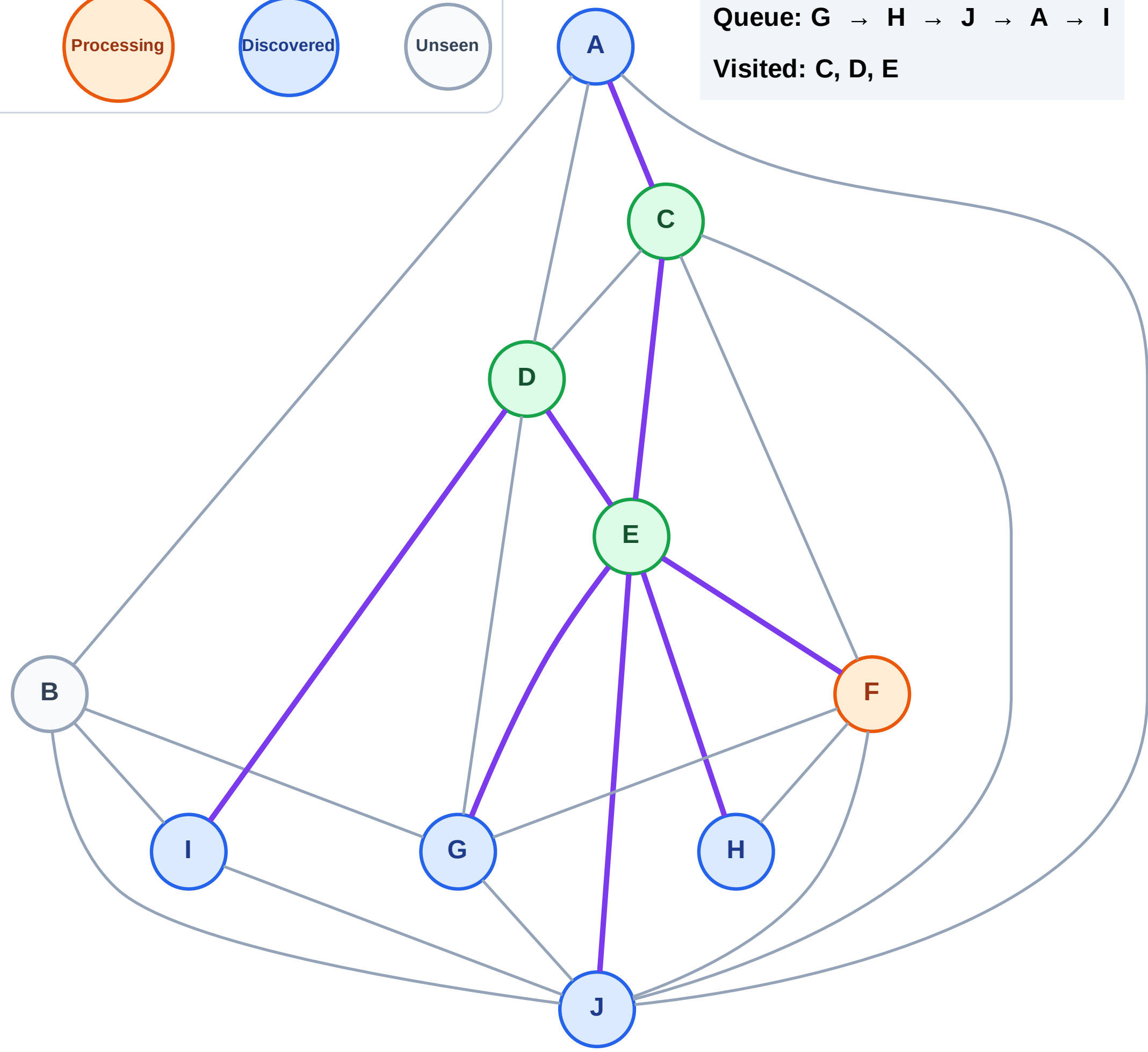
Discovered



Unseen

Queue: G → H → J → A → I

Visited: C, D, E





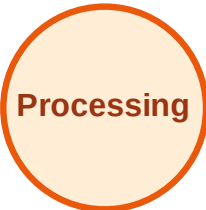
# BFS Traversal

Step 9: No new neighbors from F

## Legend



Complete



Processing



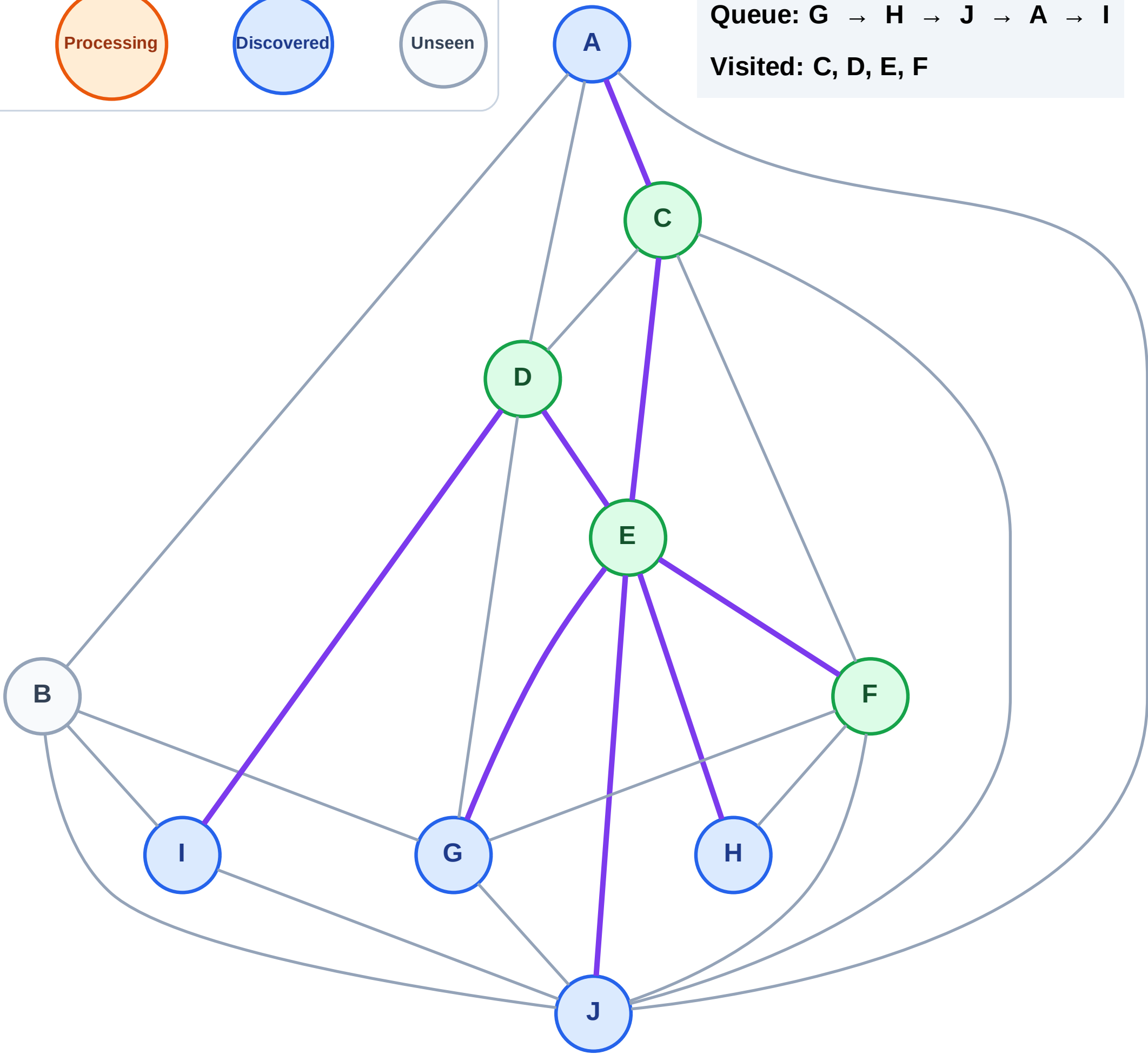
Discovered



Unseen

Queue: G → H → J → A → I

Visited: C, D, E, F



# BFS Traversal

Step 10: Process node G

## Legend



Complete



Processing



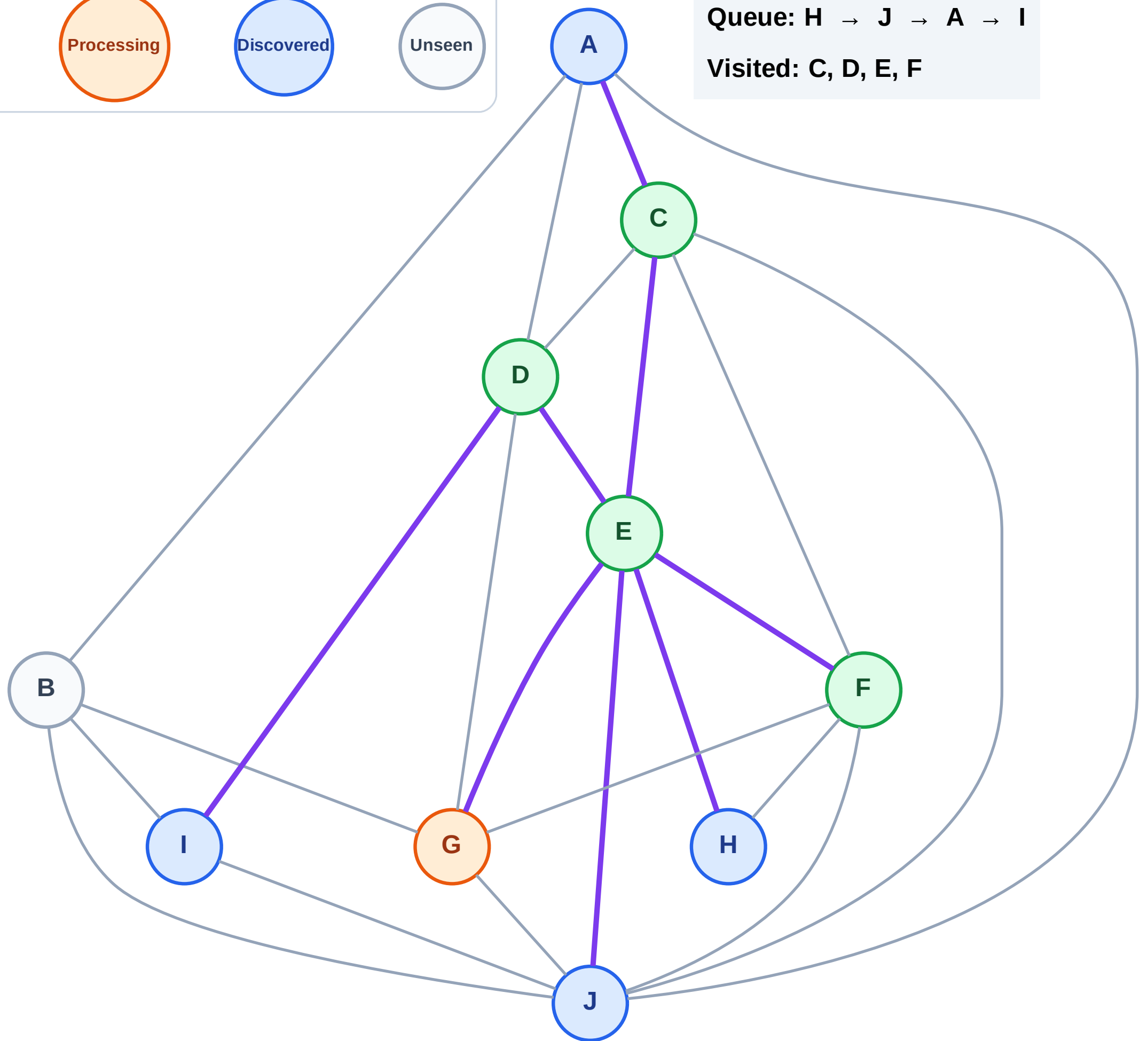
Discovered



Unseen

Queue: H → J → A → I

Visited: C, D, E, F



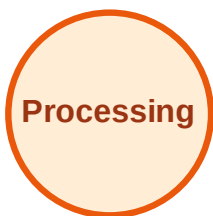
# BFS Traversal

Step 11: Discover B from G

## Legend



Complete



Processing



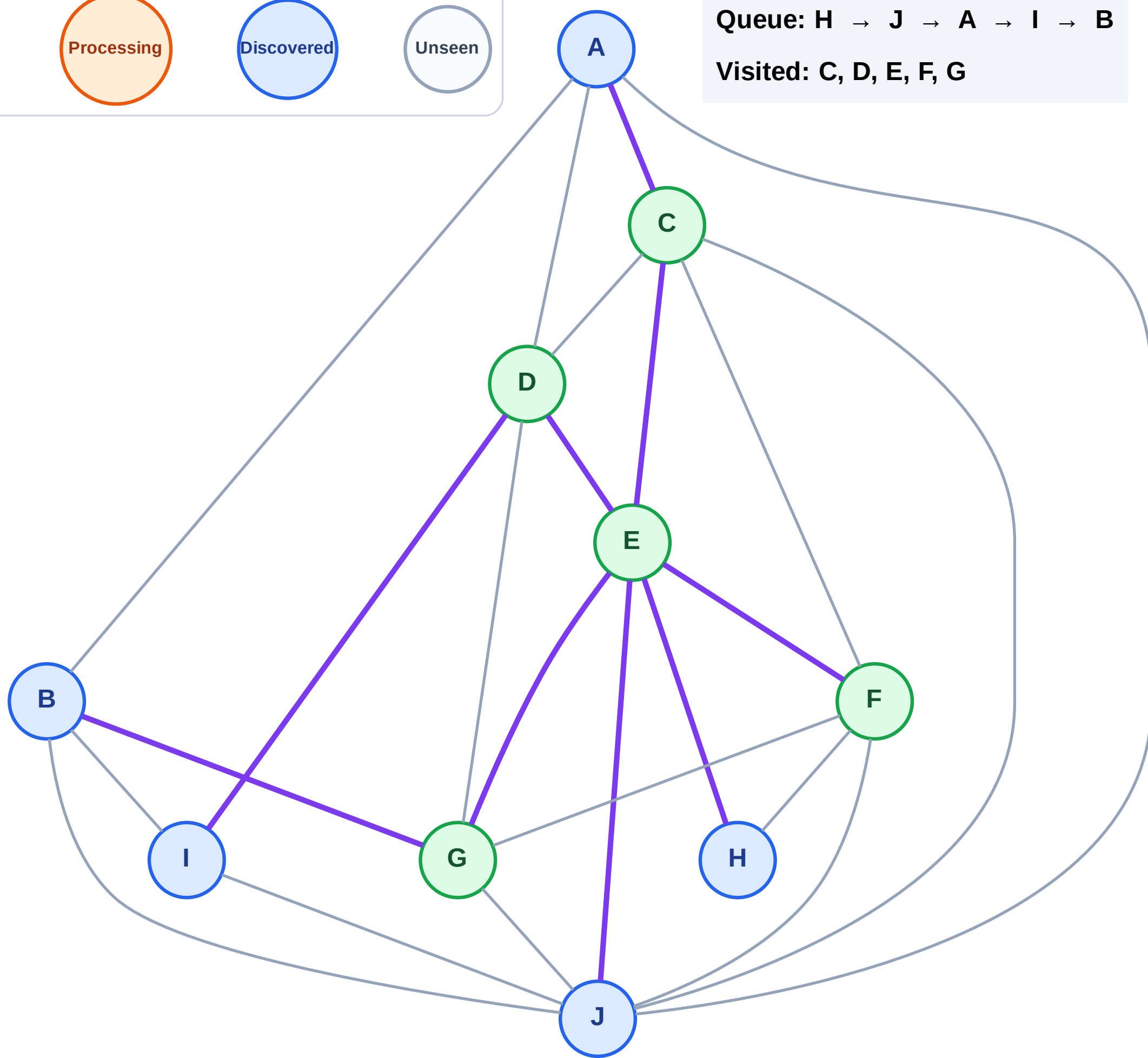
Discovered



Unseen

Queue: H → J → A → I → B

Visited: C, D, E, F, G



# BFS Traversal

Step 12: Process node H

## Legend



Complete



Processing



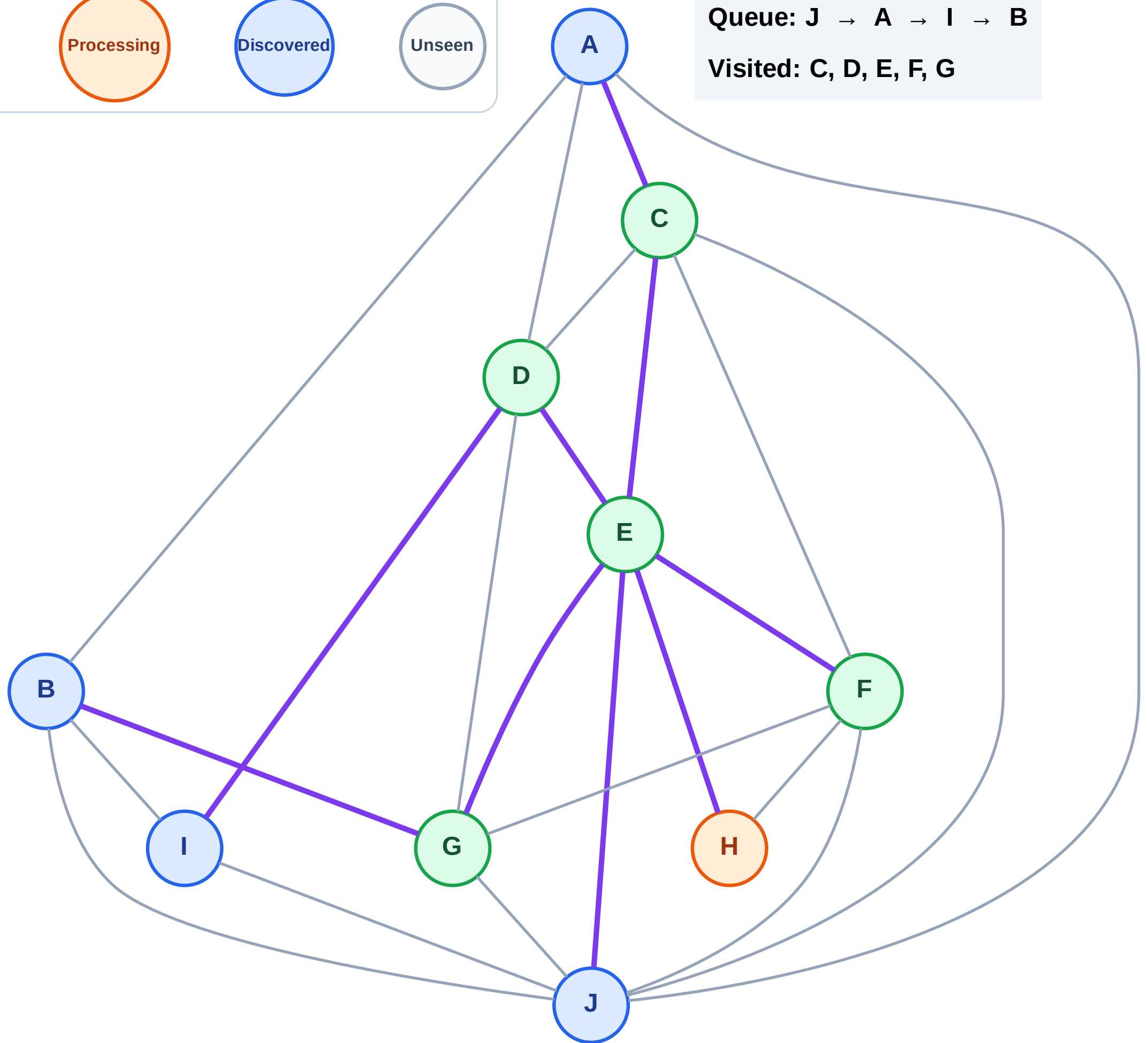
Discovered



Unseen

Queue: J → A → I → B

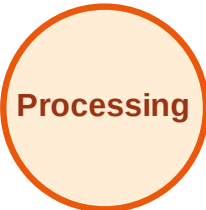
Visited: C, D, E, F, G



# BFS Traversal

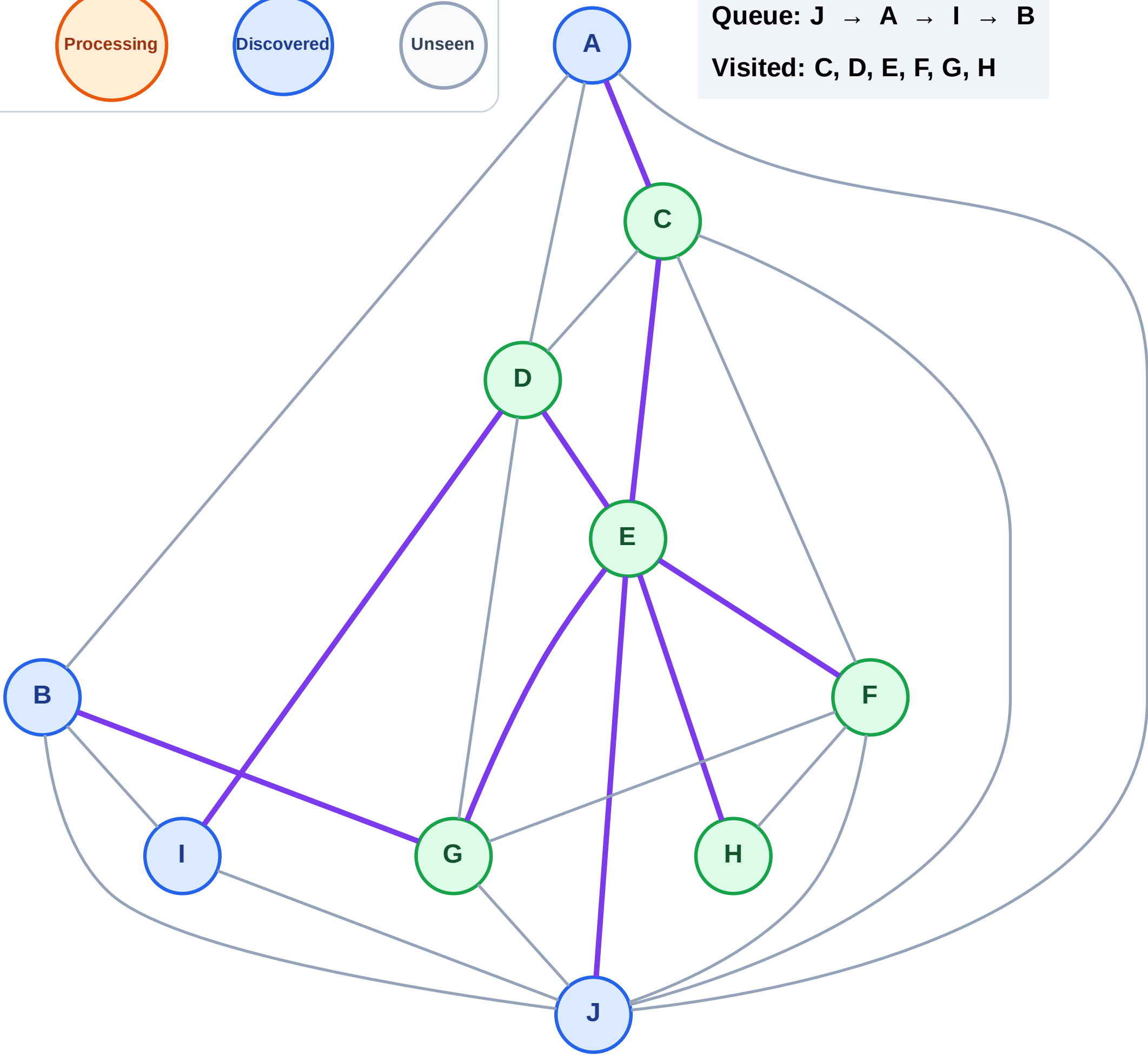
Step 13: No new neighbors from H

## Legend



Queue: J → A → I → B

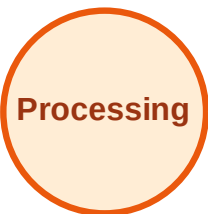
Visited: C, D, E, F, G, H



# BFS Traversal

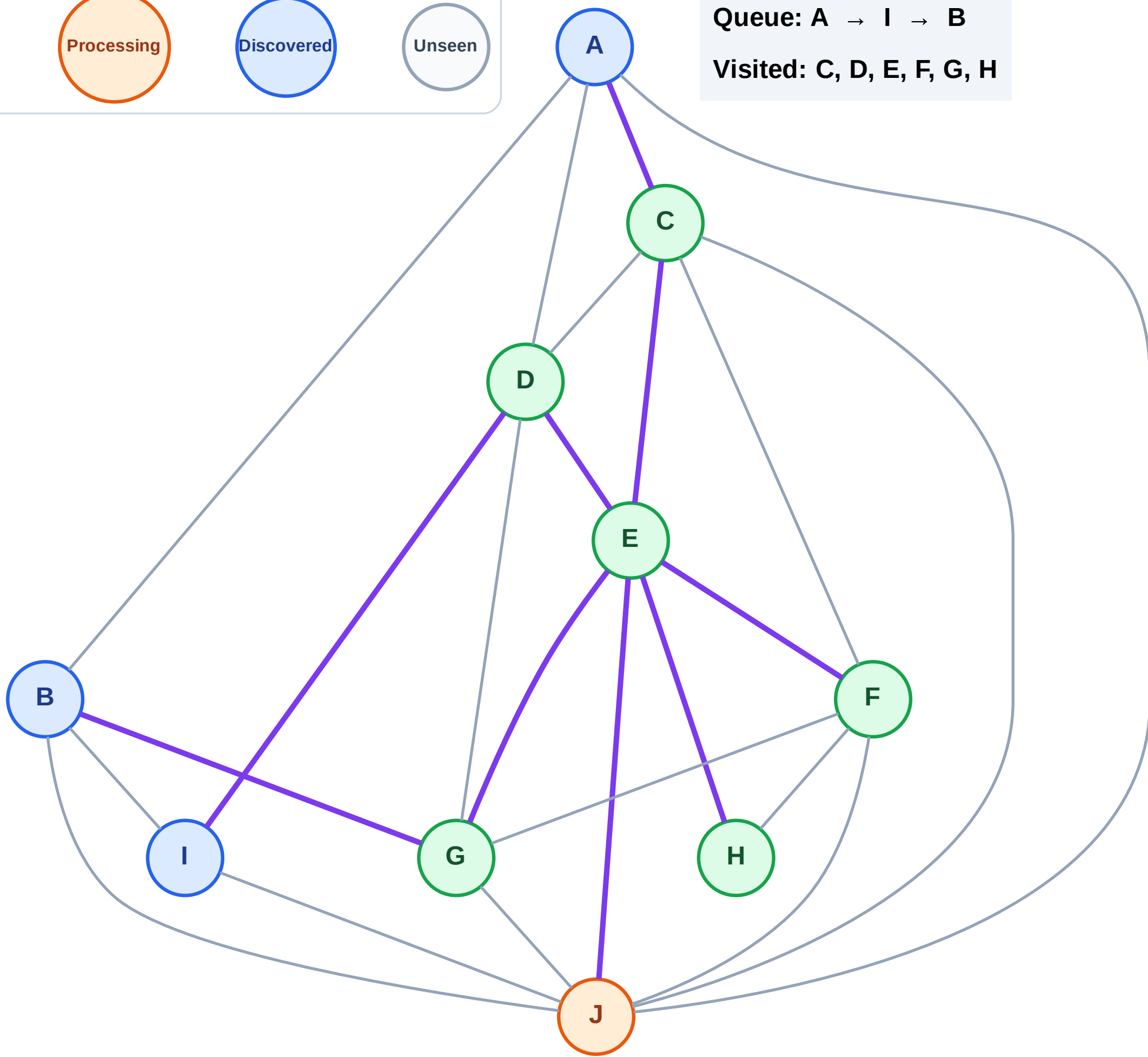
Step 14: Process node J

## Legend



Queue: A → I → B

Visited: C, D, E, F, G, H

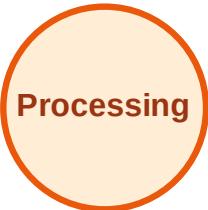




# BFS Traversal

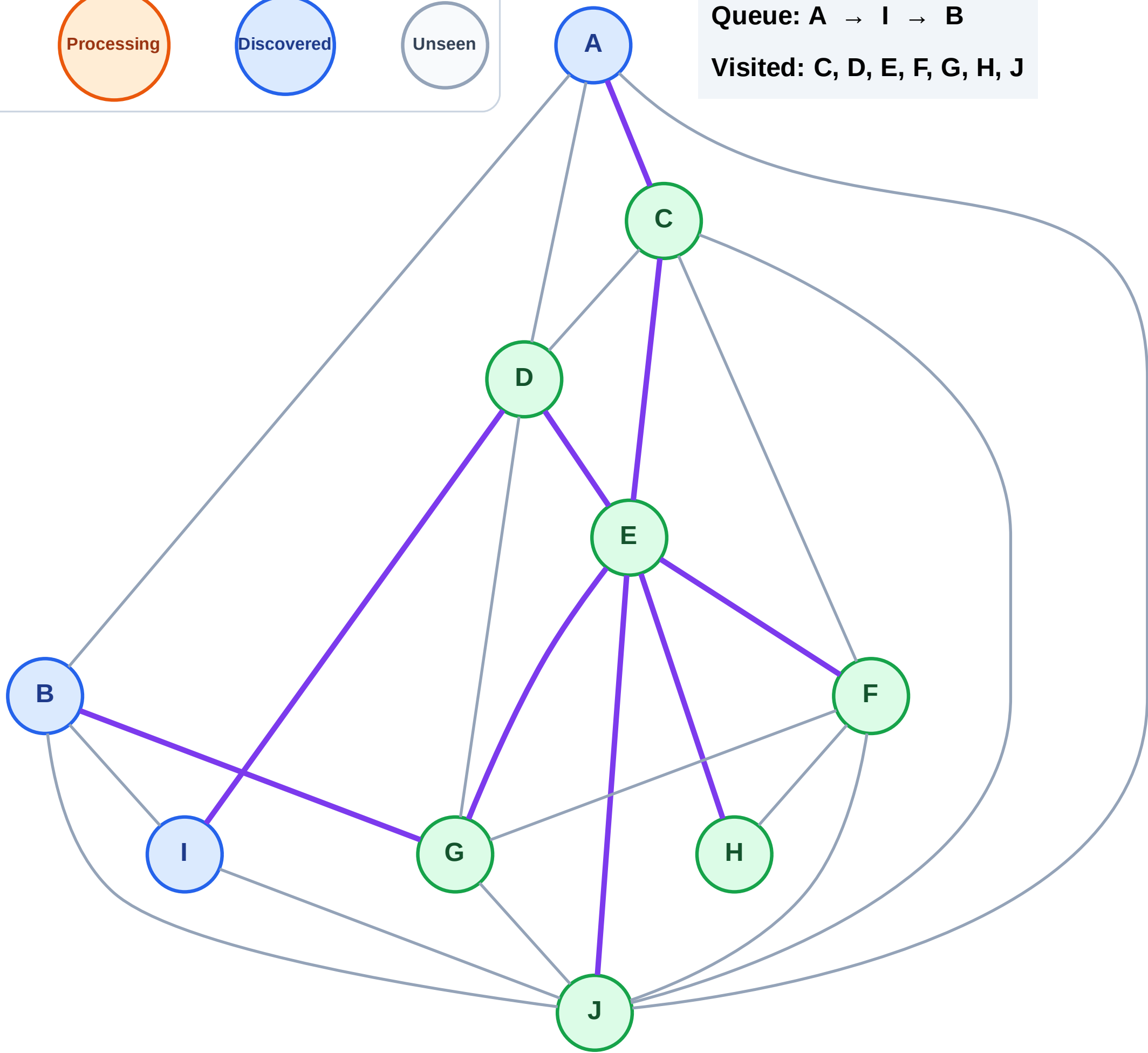
Step 15: No new neighbors from J

## Legend



Queue: A → I → B

Visited: C, D, E, F, G, H, J



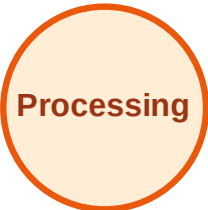
# BFS Traversal

Step 16: Process node A

## Legend



Complete



Processing



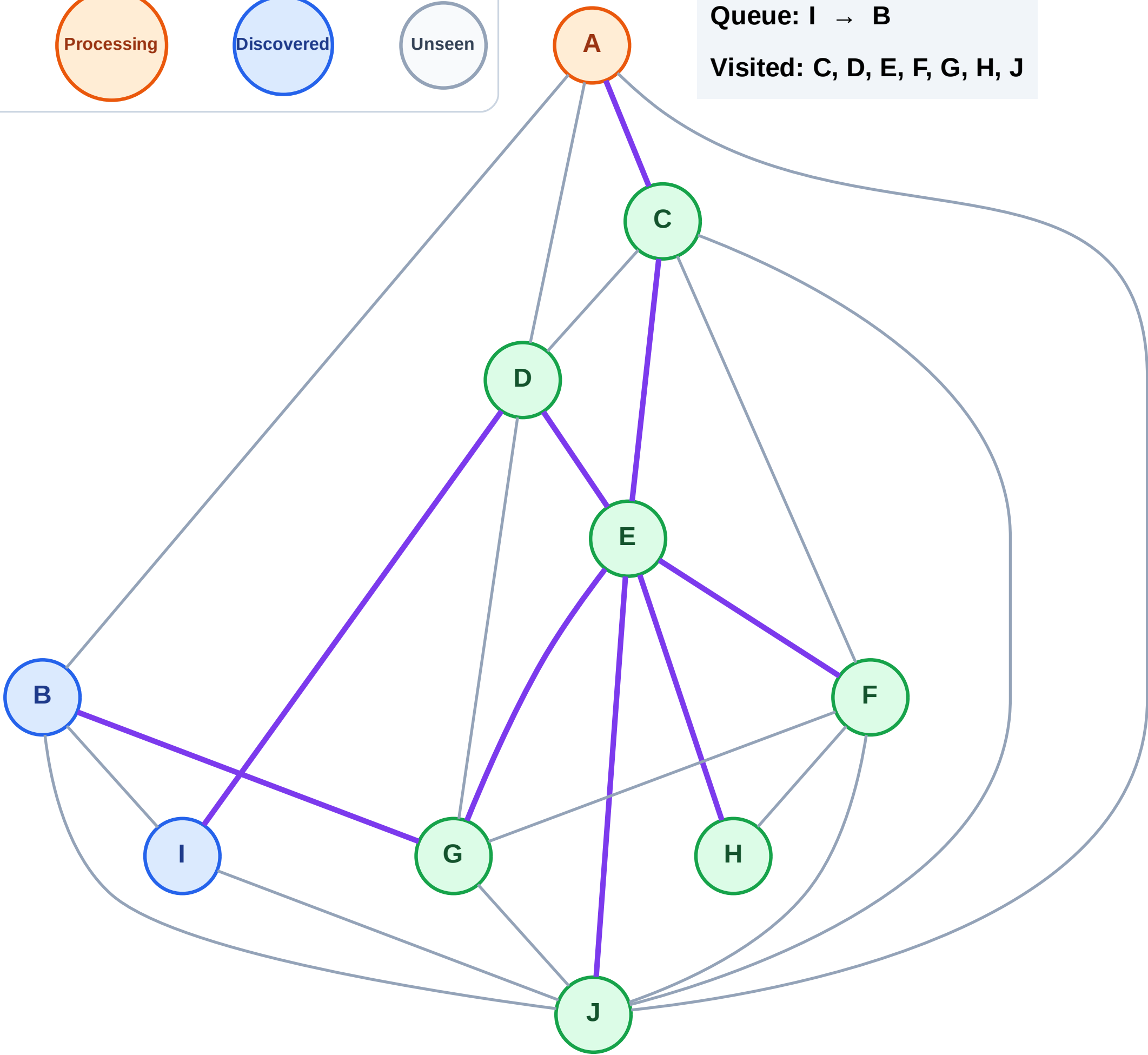
Discovered



Unseen

Queue: I → B

Visited: C, D, E, F, G, H, J

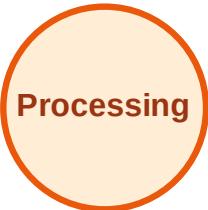




# BFS Traversal

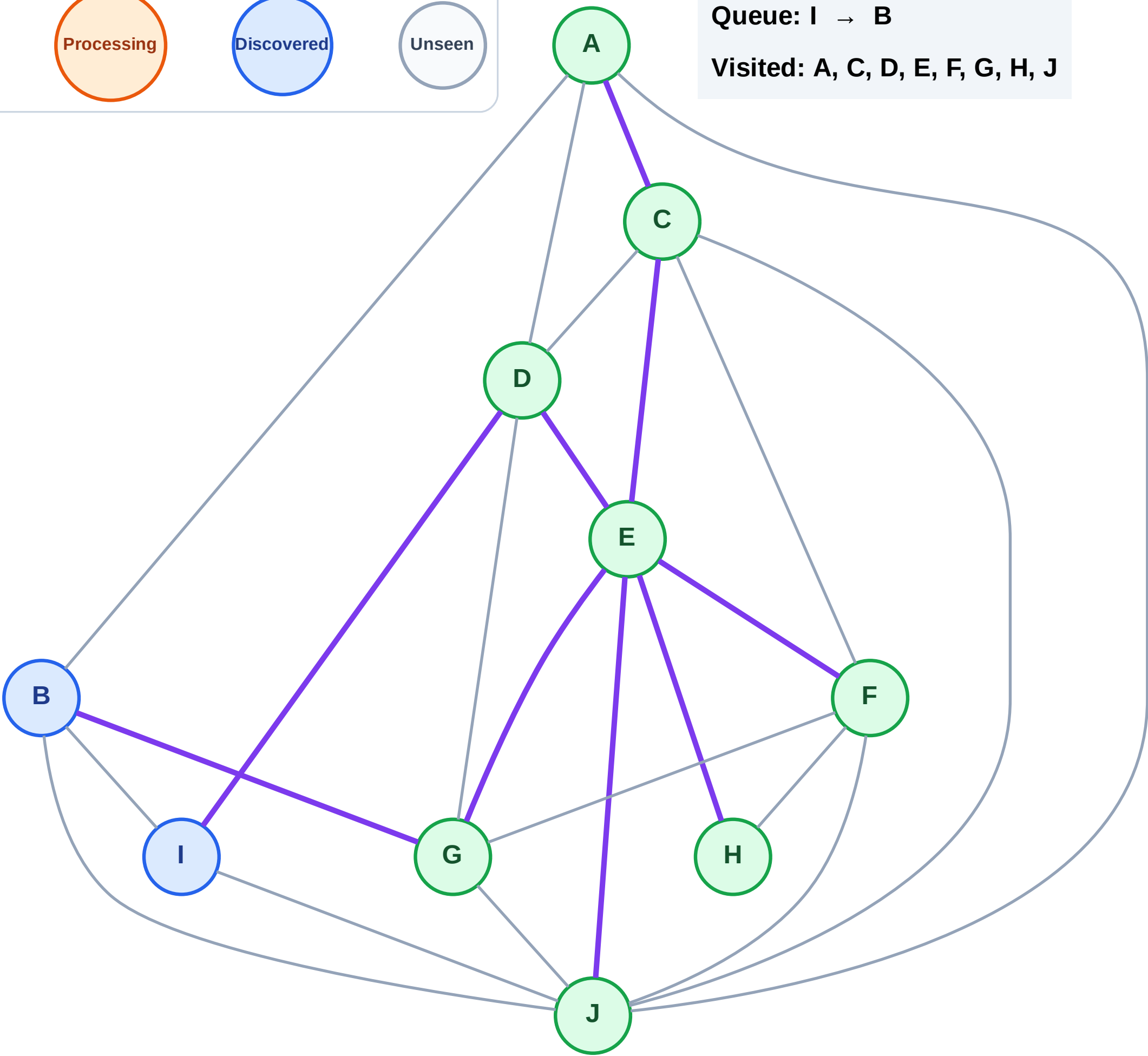
Step 17: No new neighbors from A

## Legend



Queue: I → B

Visited: A, C, D, E, F, G, H, J



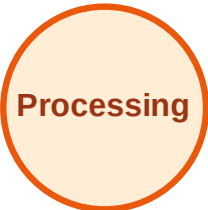
# BFS Traversal

Step 18: Process node I

## Legend



Complete



Processing



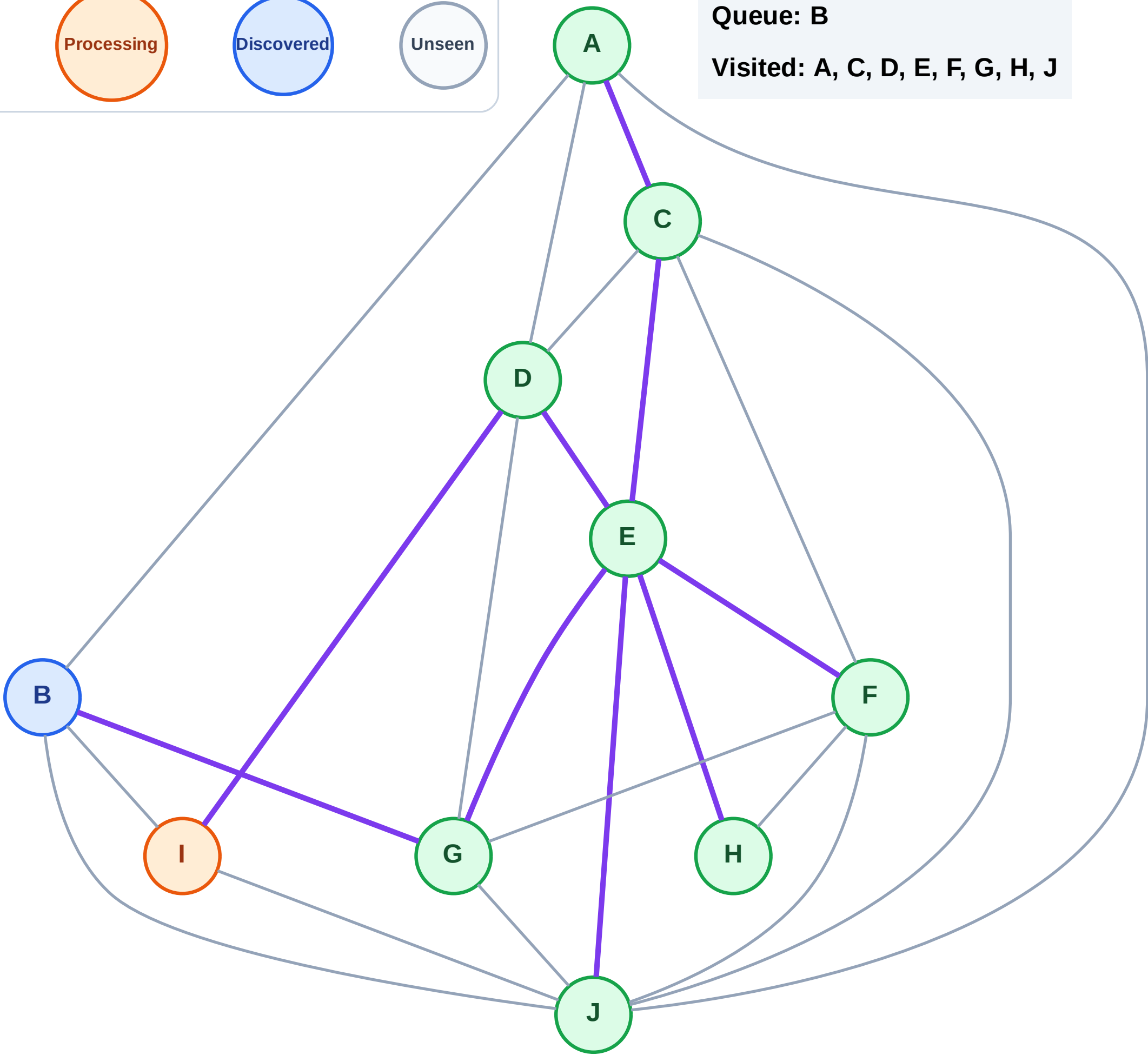
Discovered



Unseen

Queue: B

Visited: A, C, D, E, F, G, H, J



# BFS Traversal

Step 19: No new neighbors from I

Legend

Complete

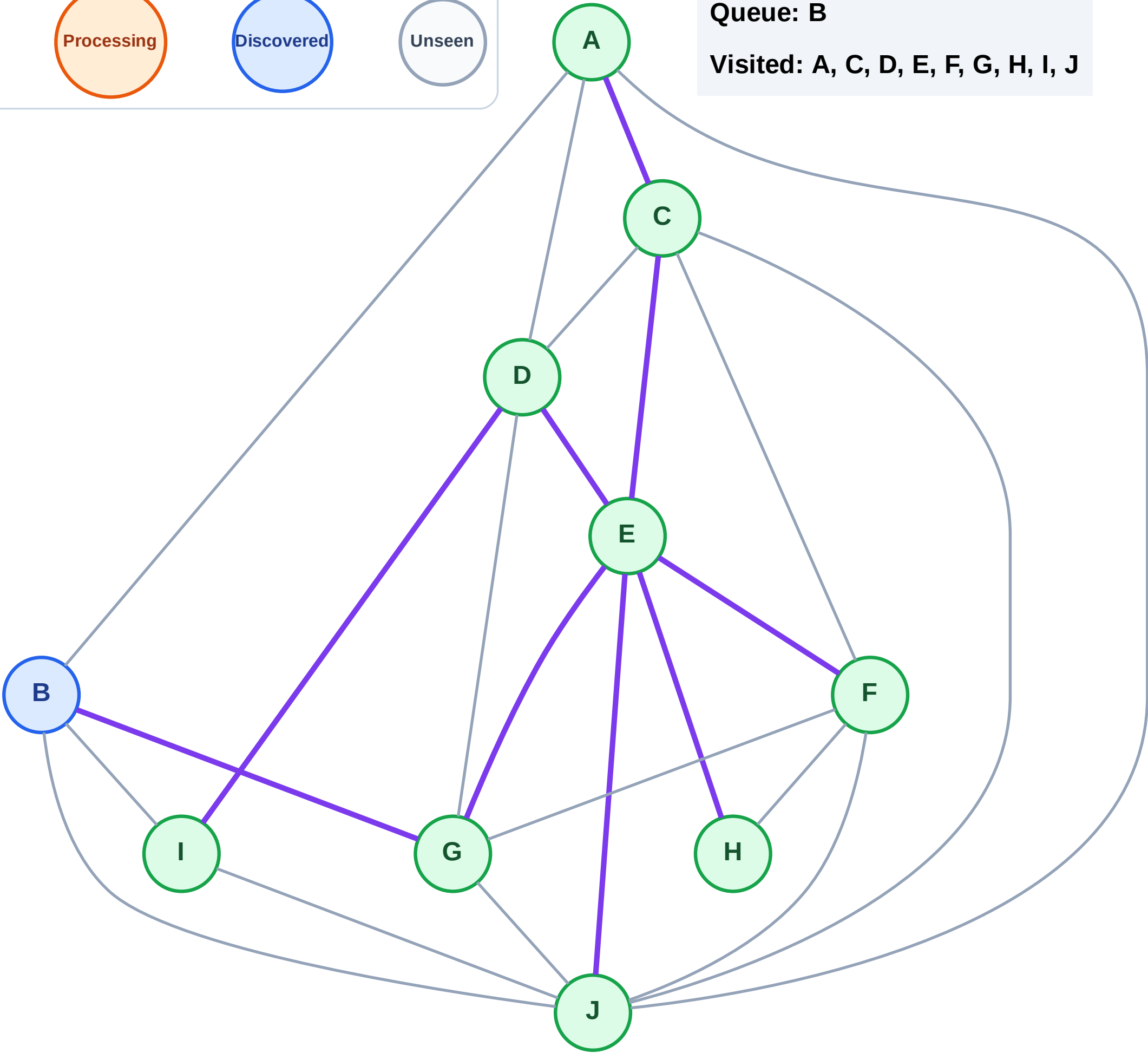
Processing

Discovered

Unseen

Queue: B

Visited: A, C, D, E, F, G, H, I, J



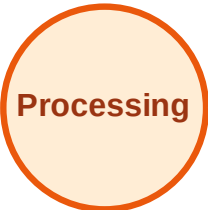
# BFS Traversal

Step 20: Process node B

## Legend



Complete



Processing



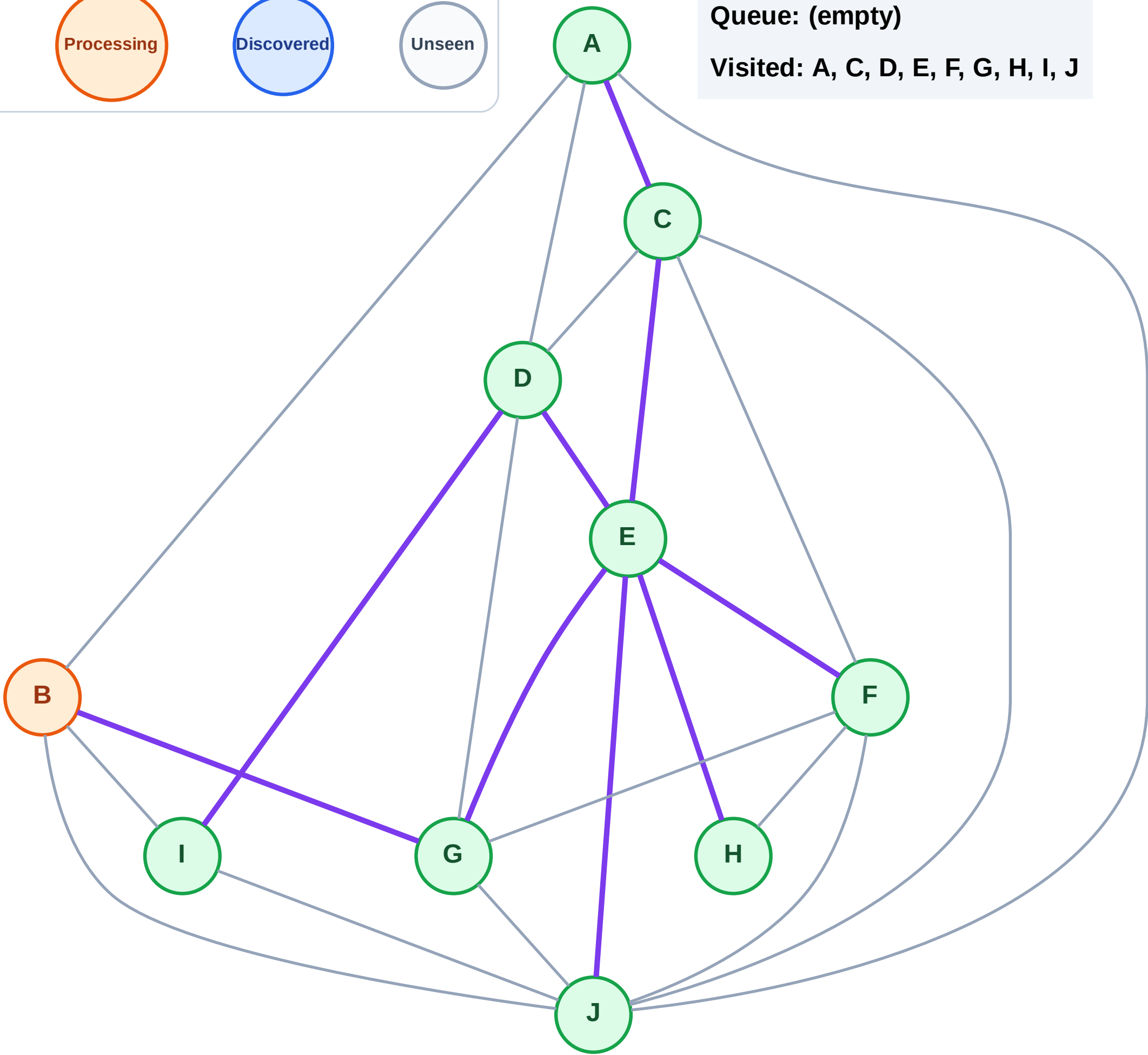
Discovered



Unseen

Queue: (empty)

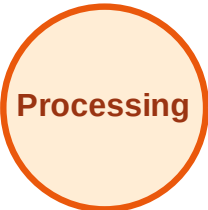
Visited: A, C, D, E, F, G, H, I, J



# BFS Traversal

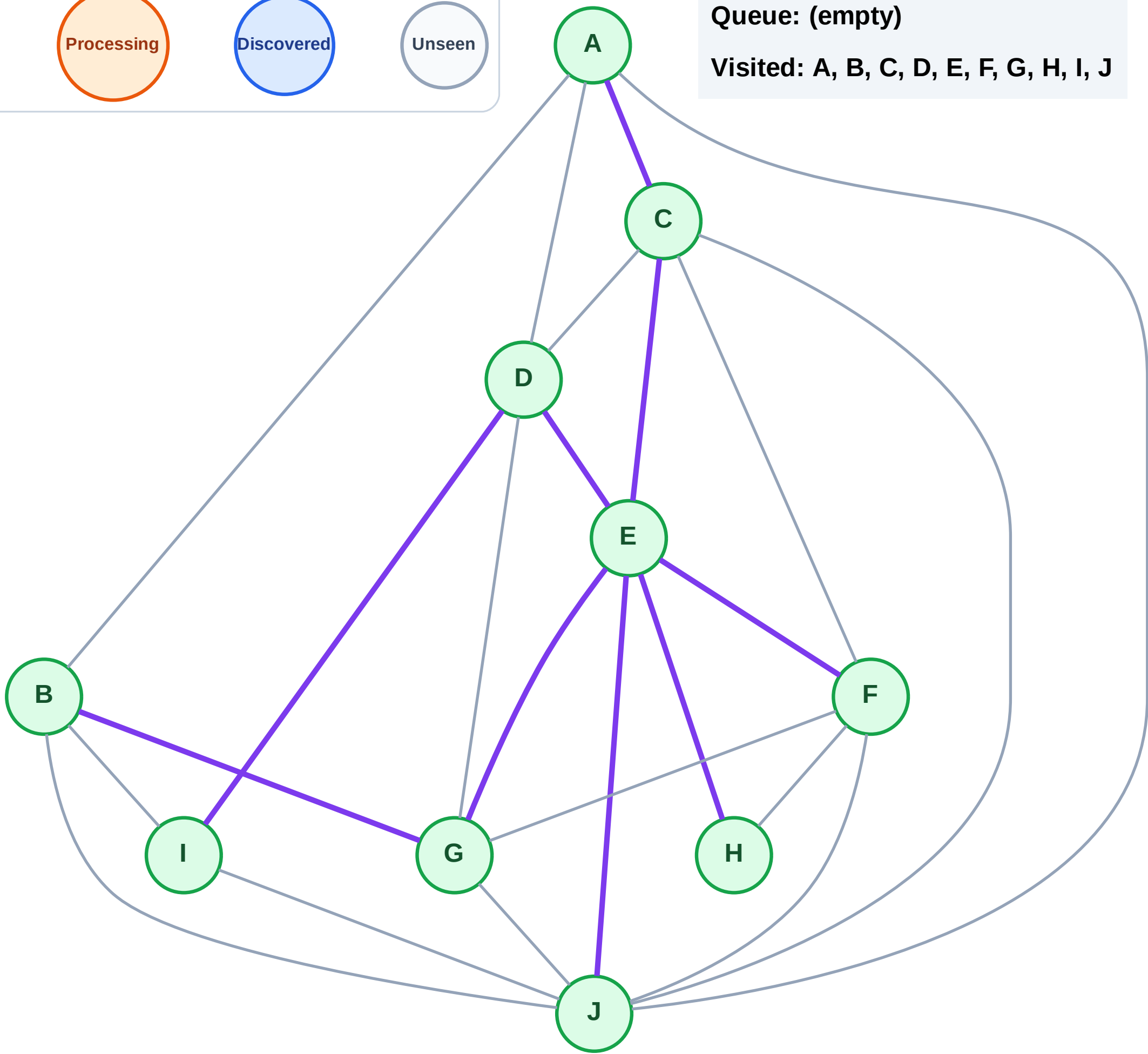
Step 21: No new neighbors from B

## Legend



Queue: (empty)

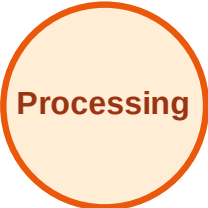
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

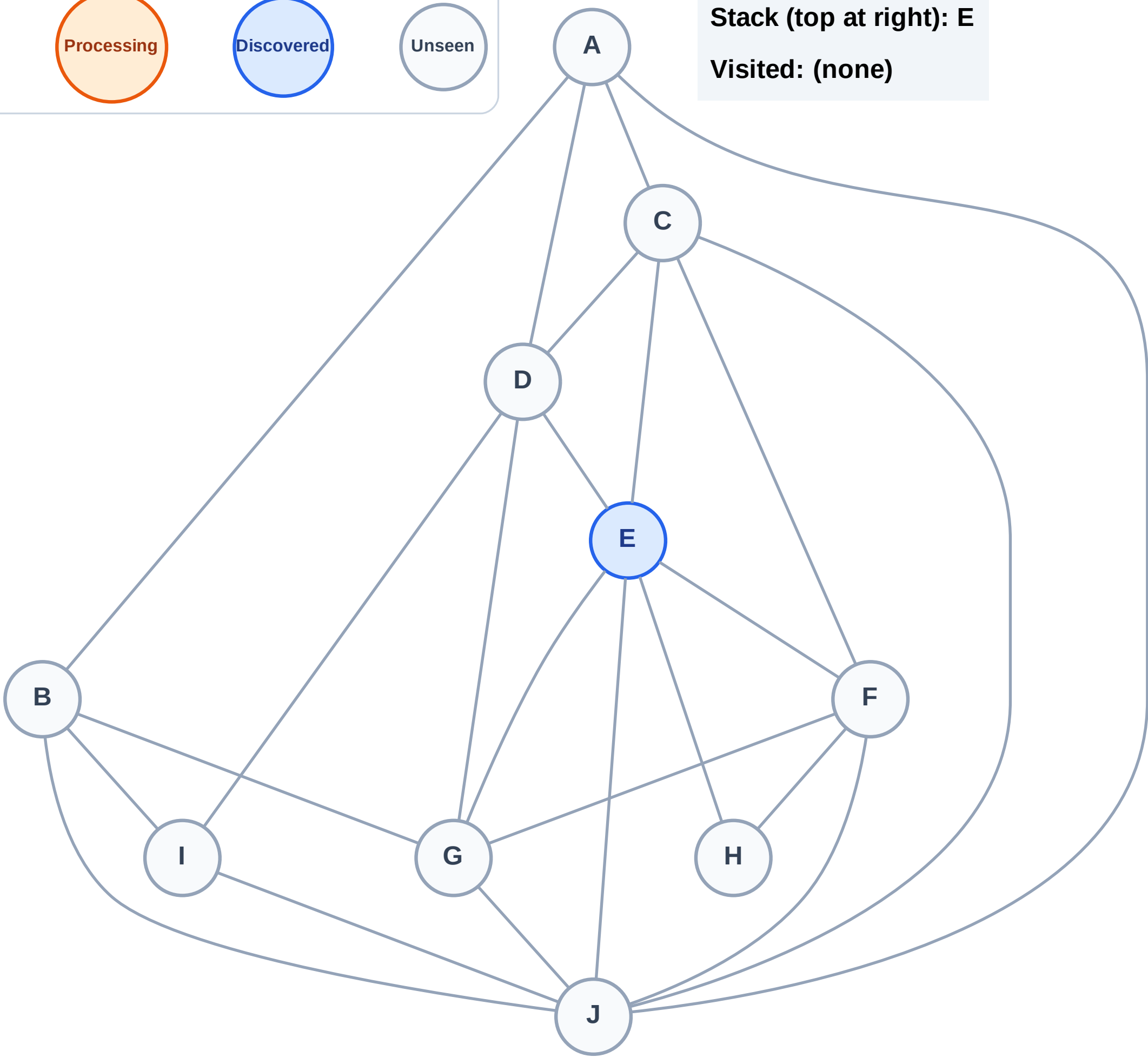
Step 1: Start at E

## Legend



Stack (top at right): E

Visited: (none)

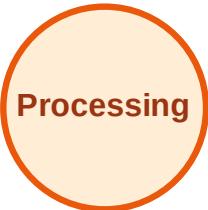




# DFS Traversal

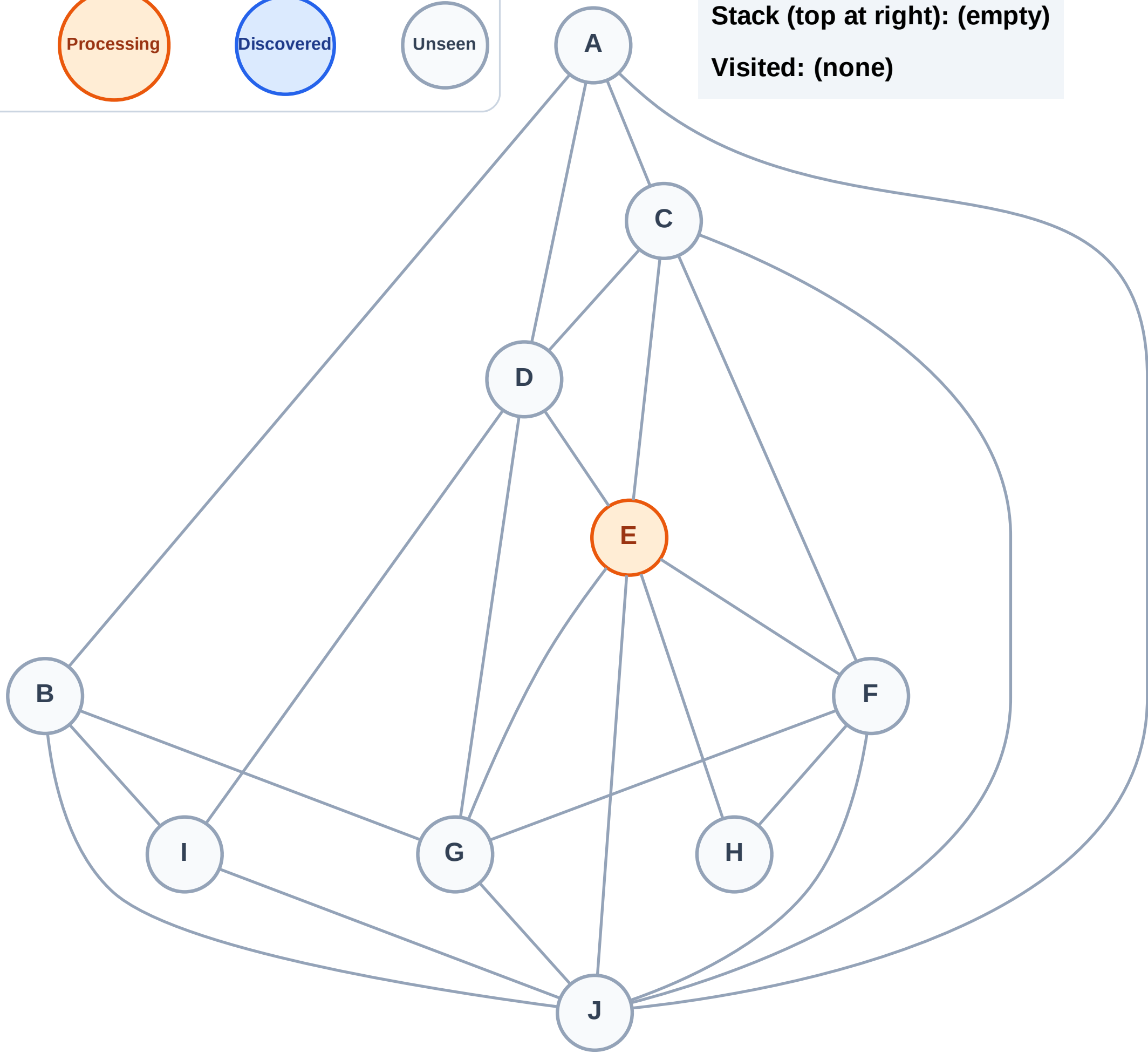
Step 2: Process node E

## Legend



Stack (top at right): (empty)

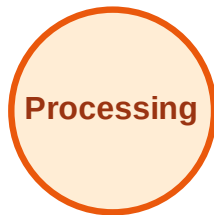
Visited: (none)



# DFS Traversal

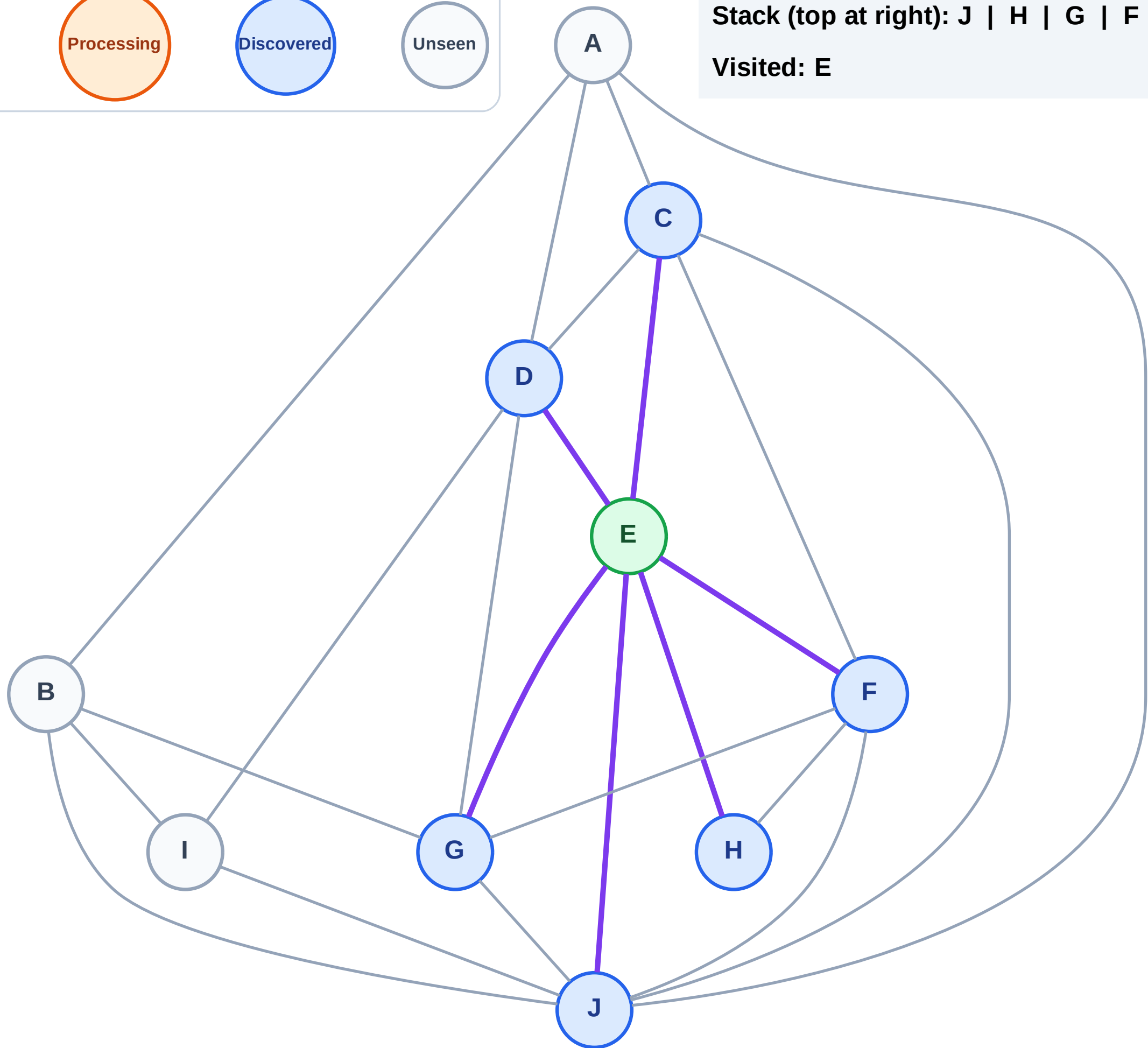
Step 3: Discover J, H, G, F, D, C from E

## Legend



Stack (top at right): J | H | G | F | D | C

Visited: E

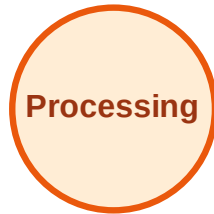




# DFS Traversal

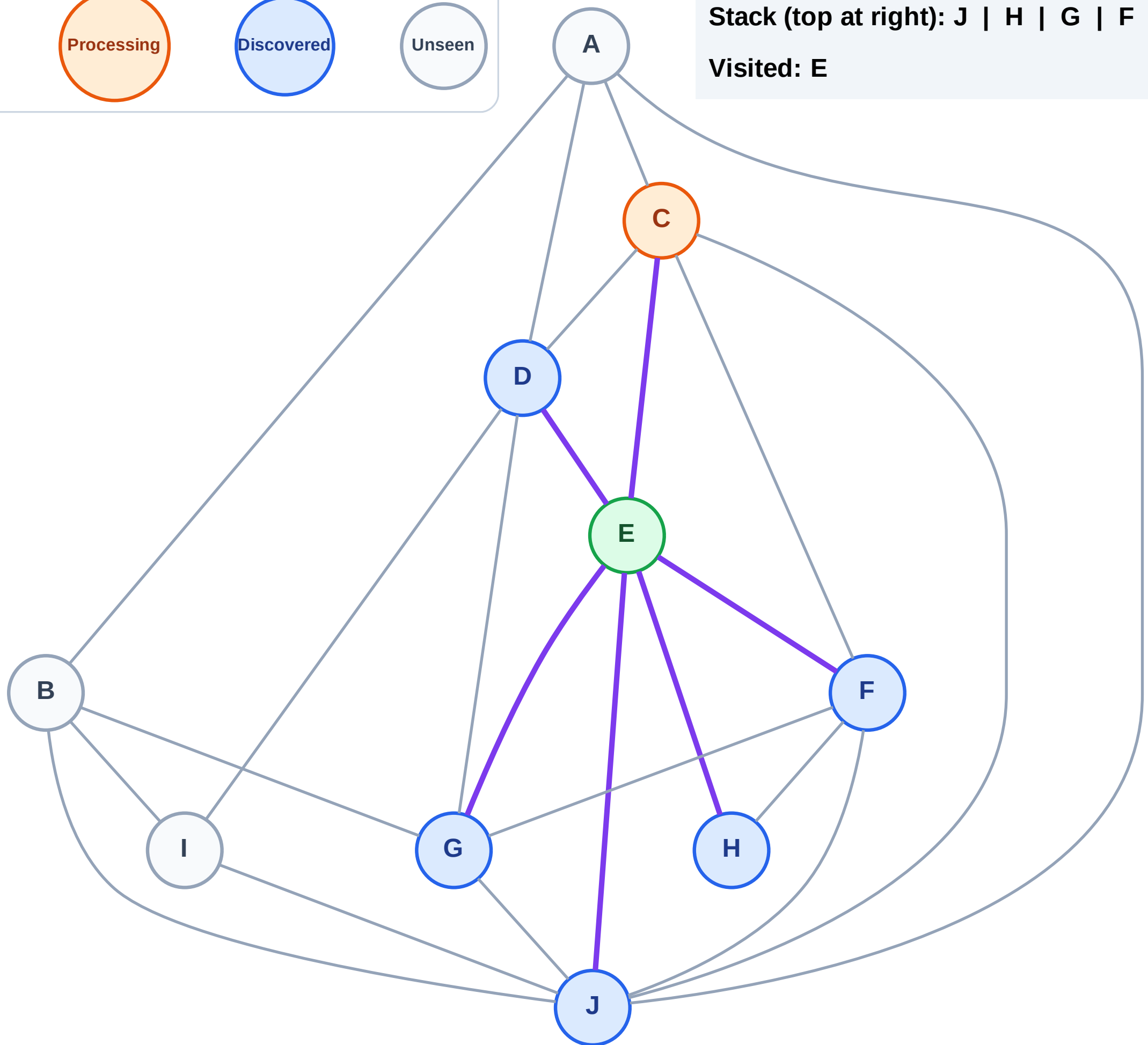
Step 4: Process node C

## Legend



Stack (top at right): J | H | G | F | D

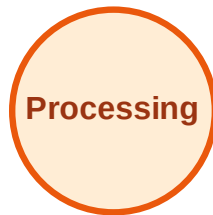
Visited: E



# DFS Traversal

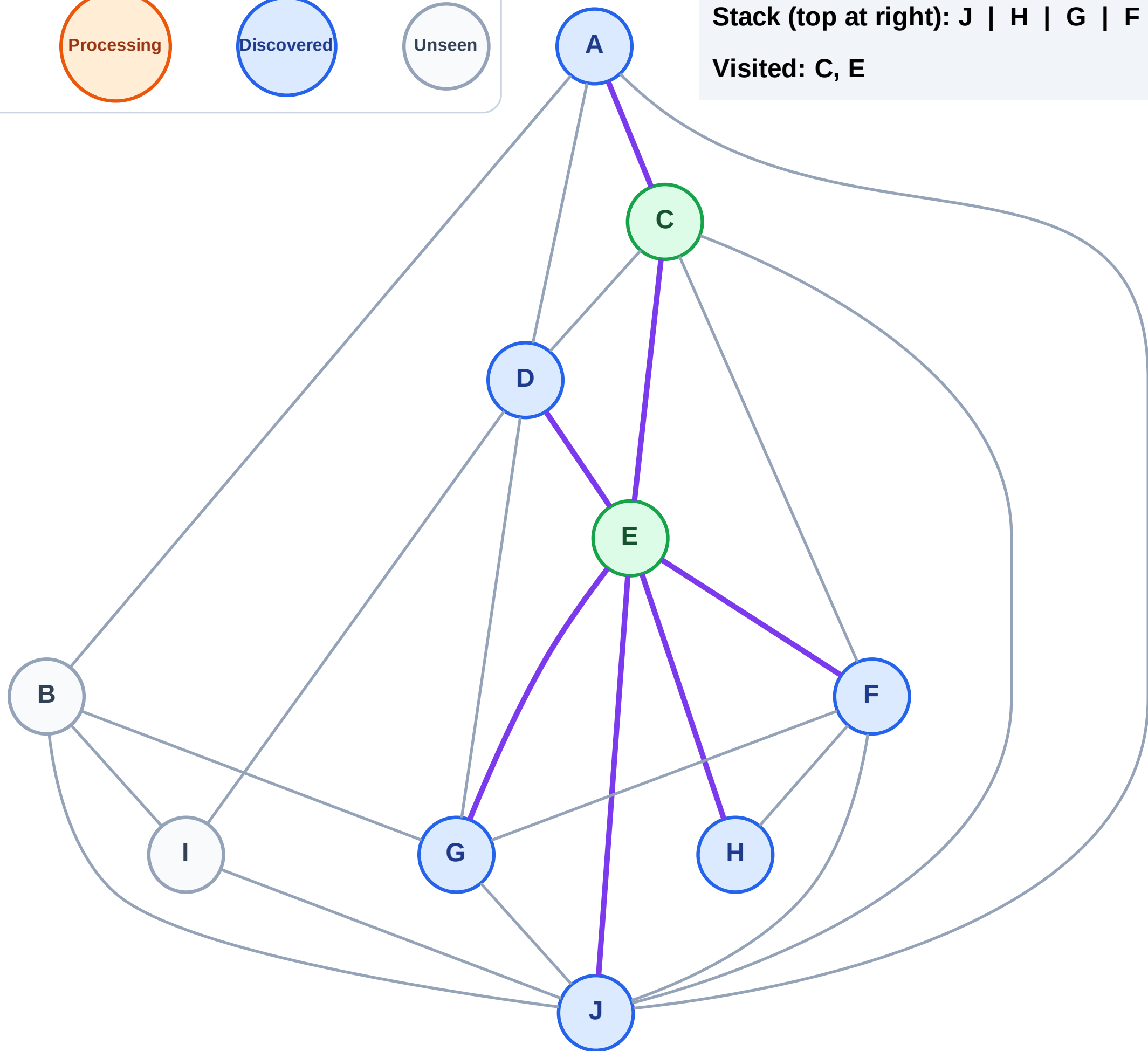
Step 5: Discover A from C

## Legend



Stack (top at right): J | H | G | F | D | A

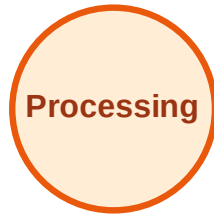
Visited: C, E



# DFS Traversal

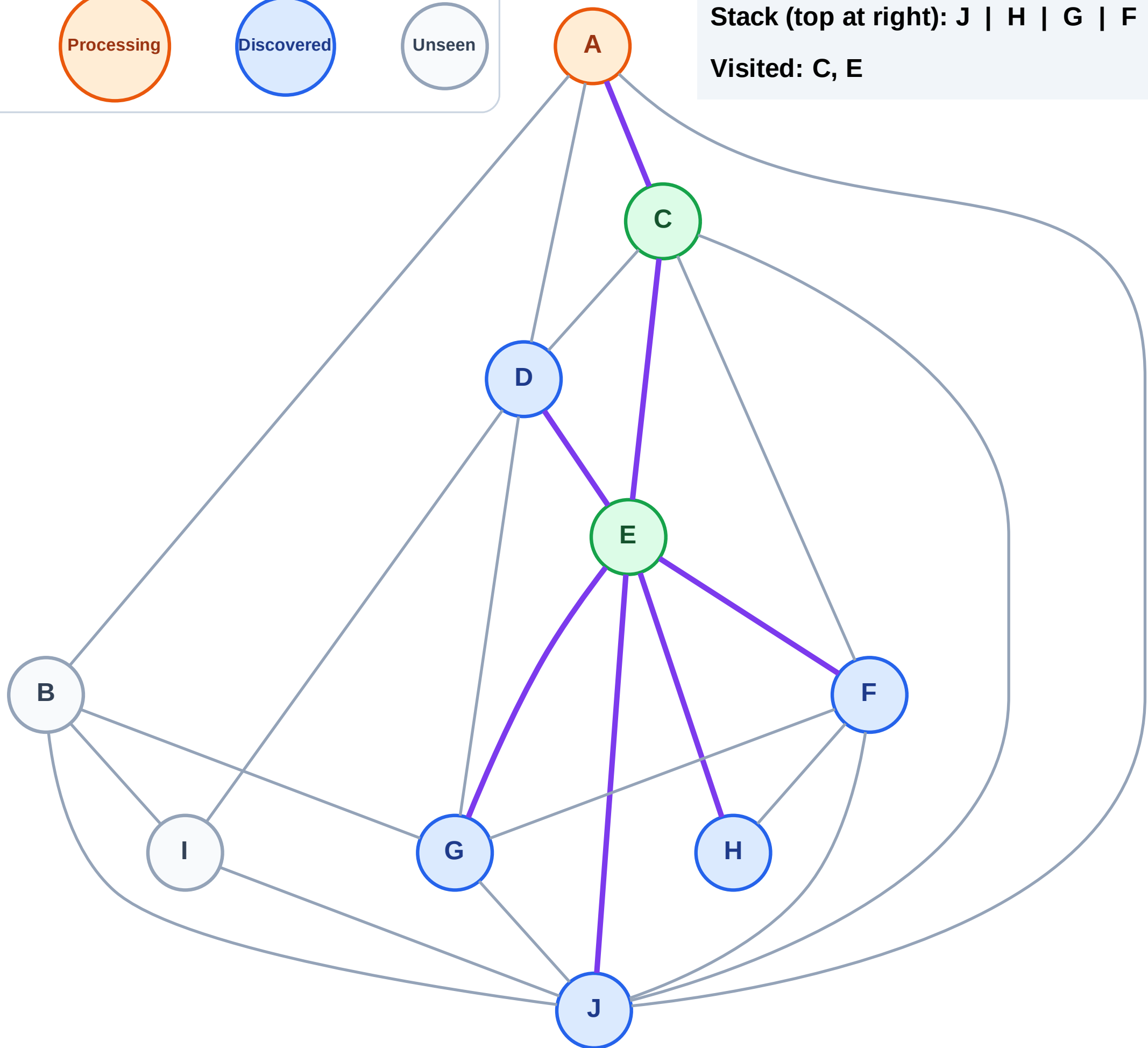
Step 6: Process node A

## Legend



Stack (top at right): J | H | G | F | D

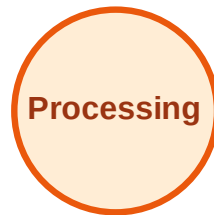
Visited: C, E



# DFS Traversal

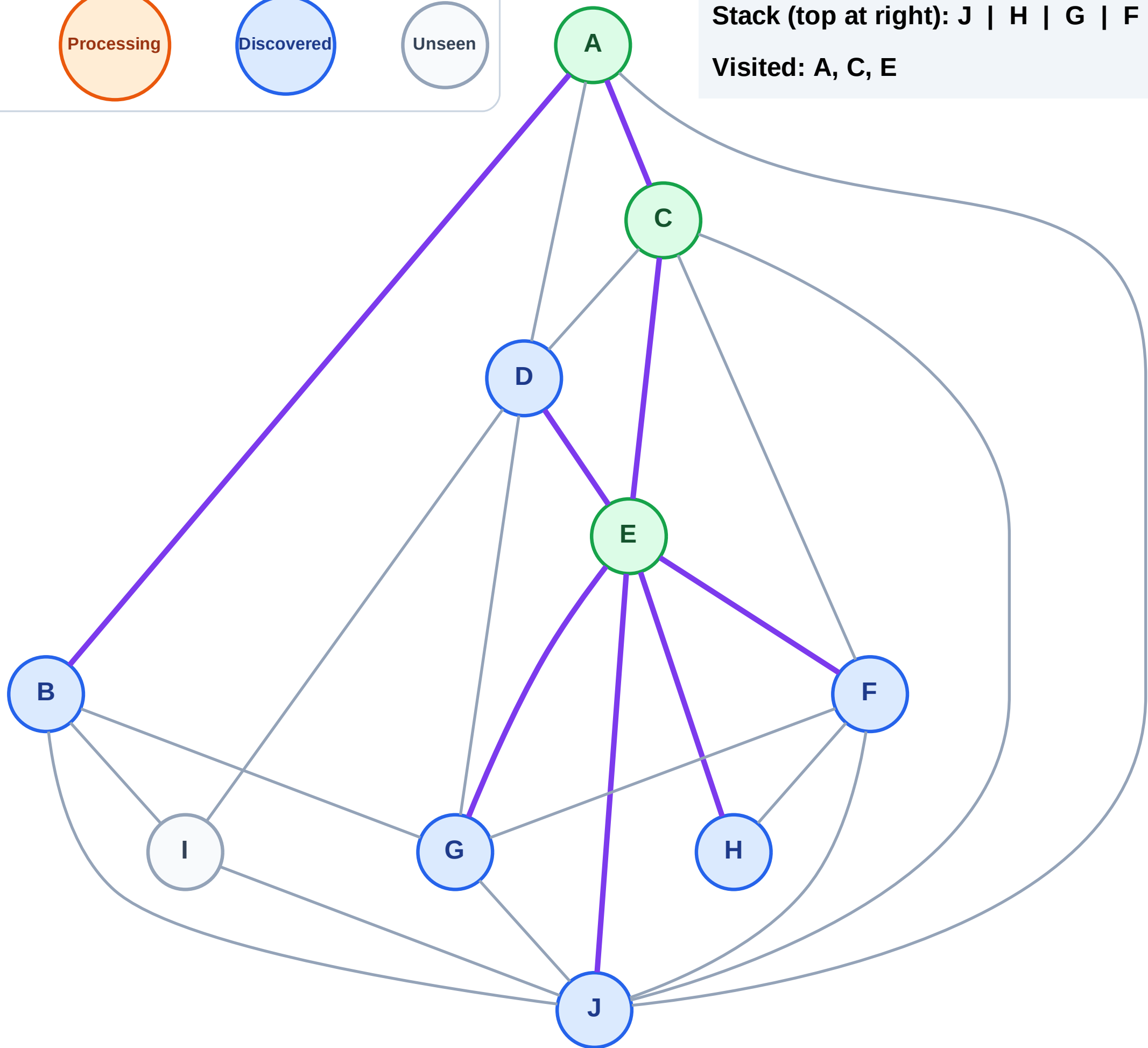
Step 7: Discover B from A

## Legend



Stack (top at right): J | H | G | F | D | B

Visited: A, C, E



# DFS Traversal

Step 8: Process node B

## Legend

Complete

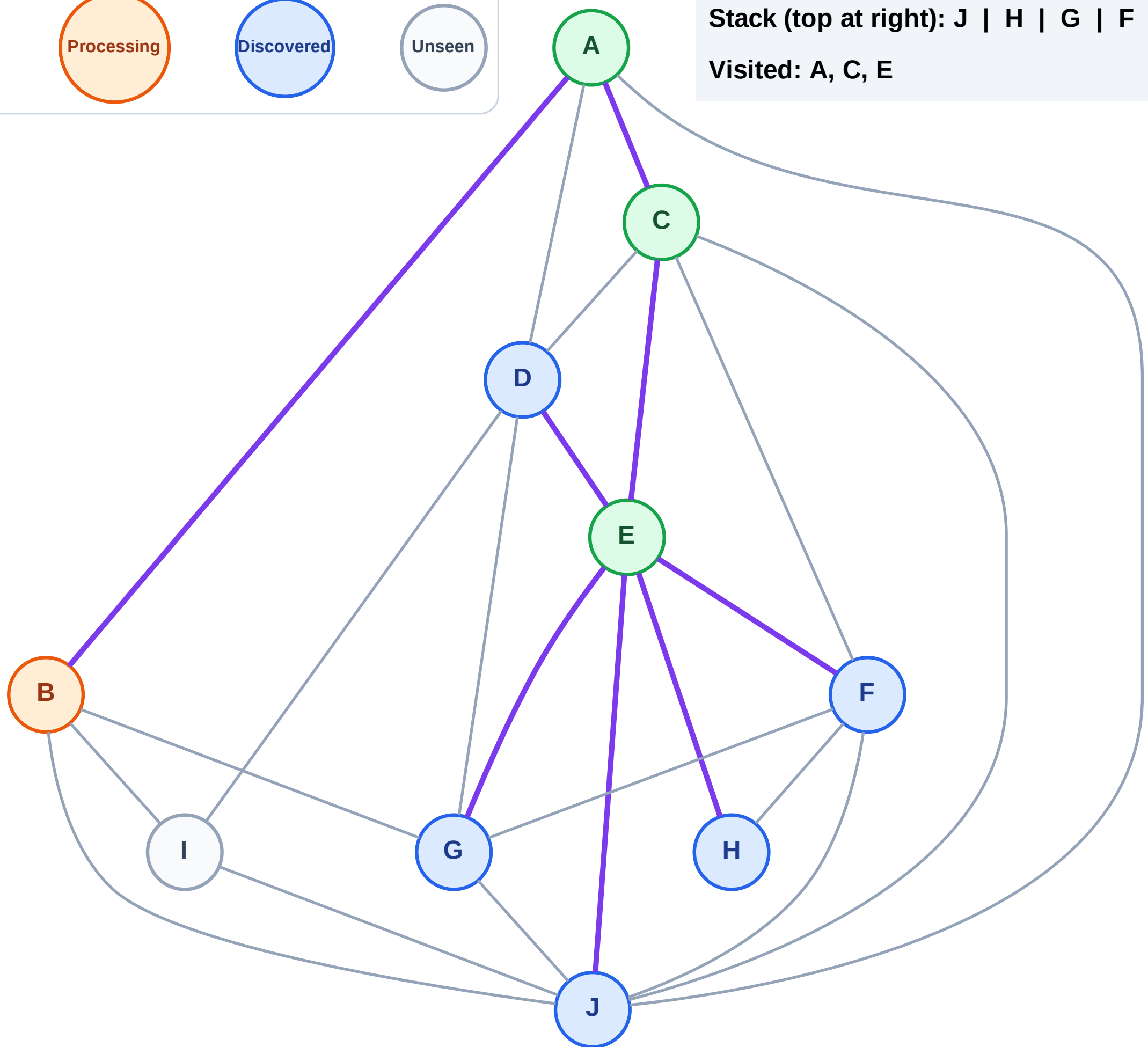
Processing

Discovered

Unseen

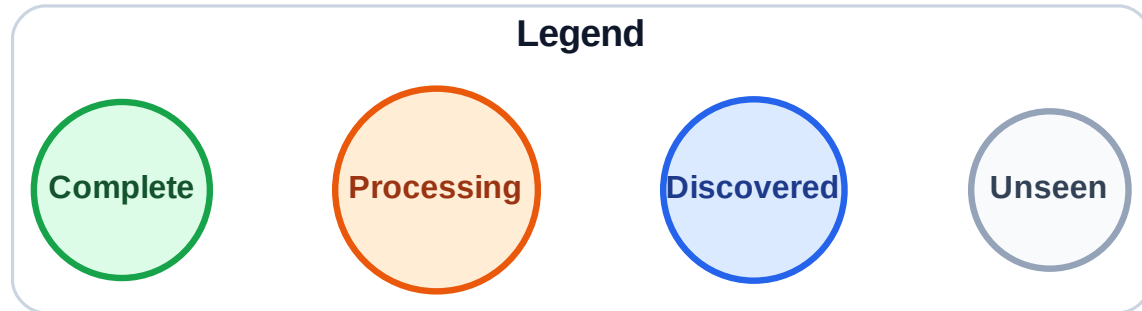
Stack (top at right): J | H | G | F | D

Visited: A, C, E



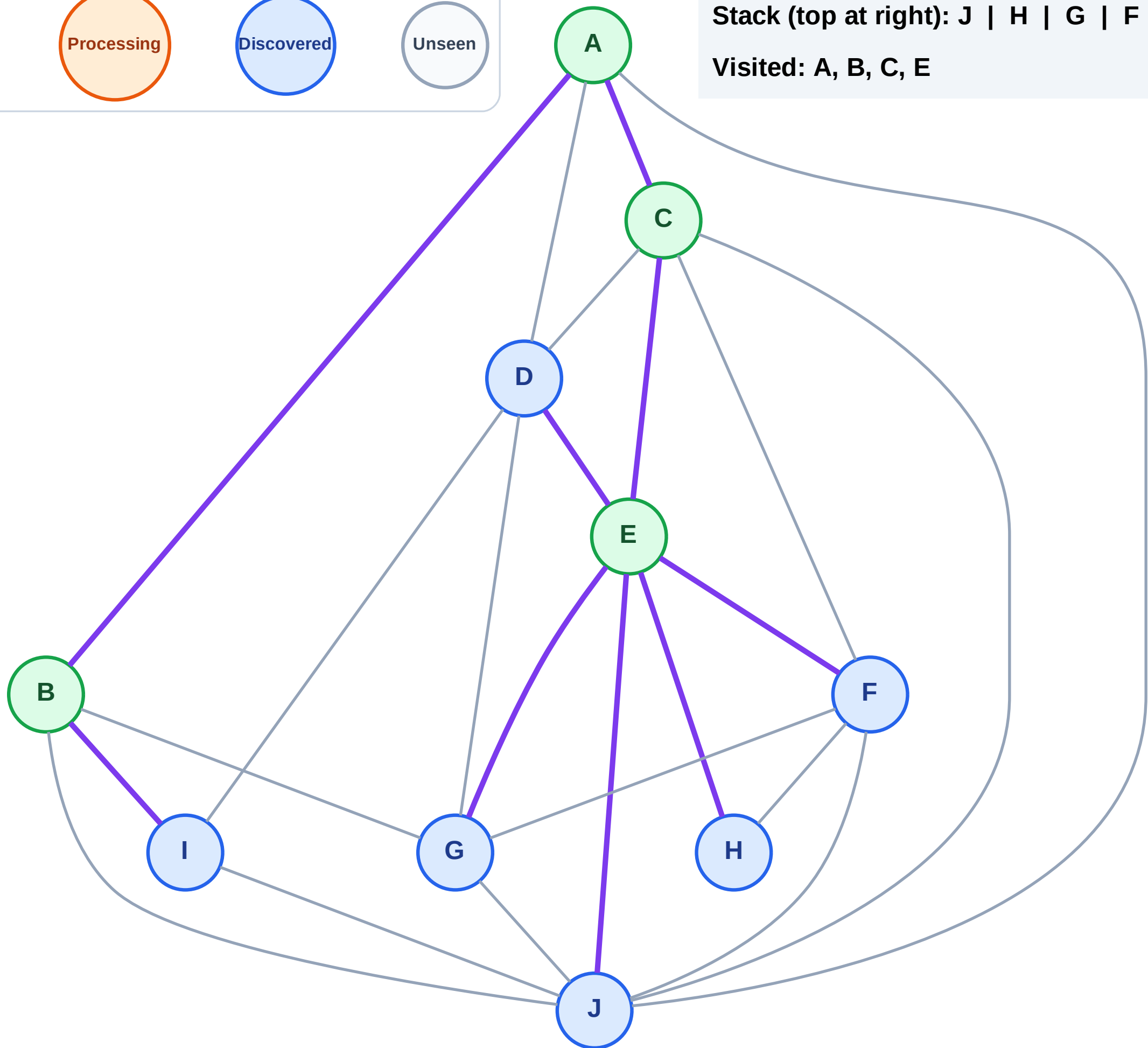
# DFS Traversal

Step 9: Discover I from B



Stack (top at right): J | H | G | F | D | I

Visited: A, B, C, E





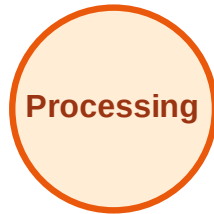
# DFS Traversal

Step 10: Process node I

## Legend



Complete



Processing



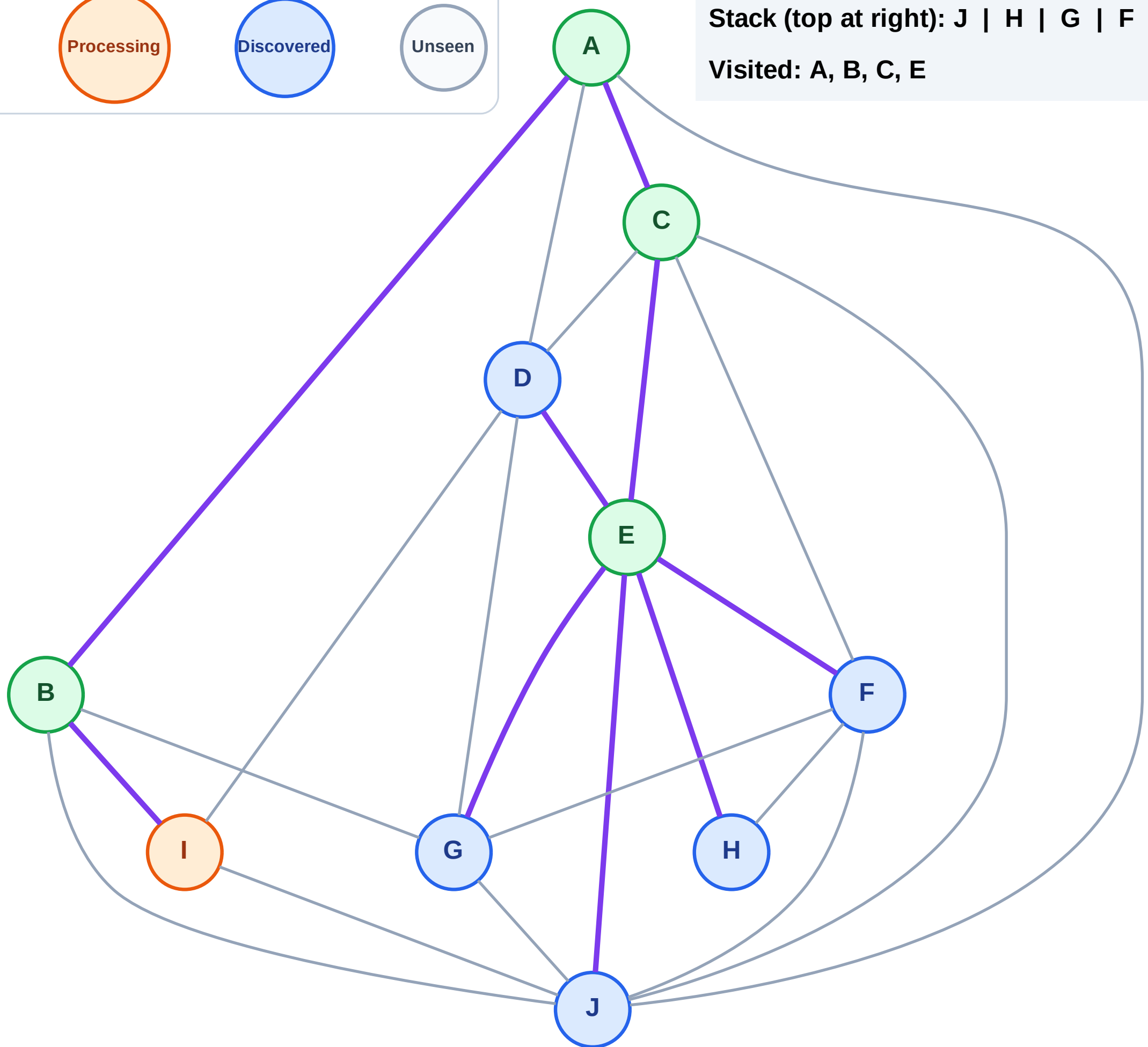
Discovered



Unseen

Stack (top at right): J | H | G | F | D

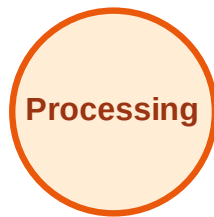
Visited: A, B, C, E



# DFS Traversal

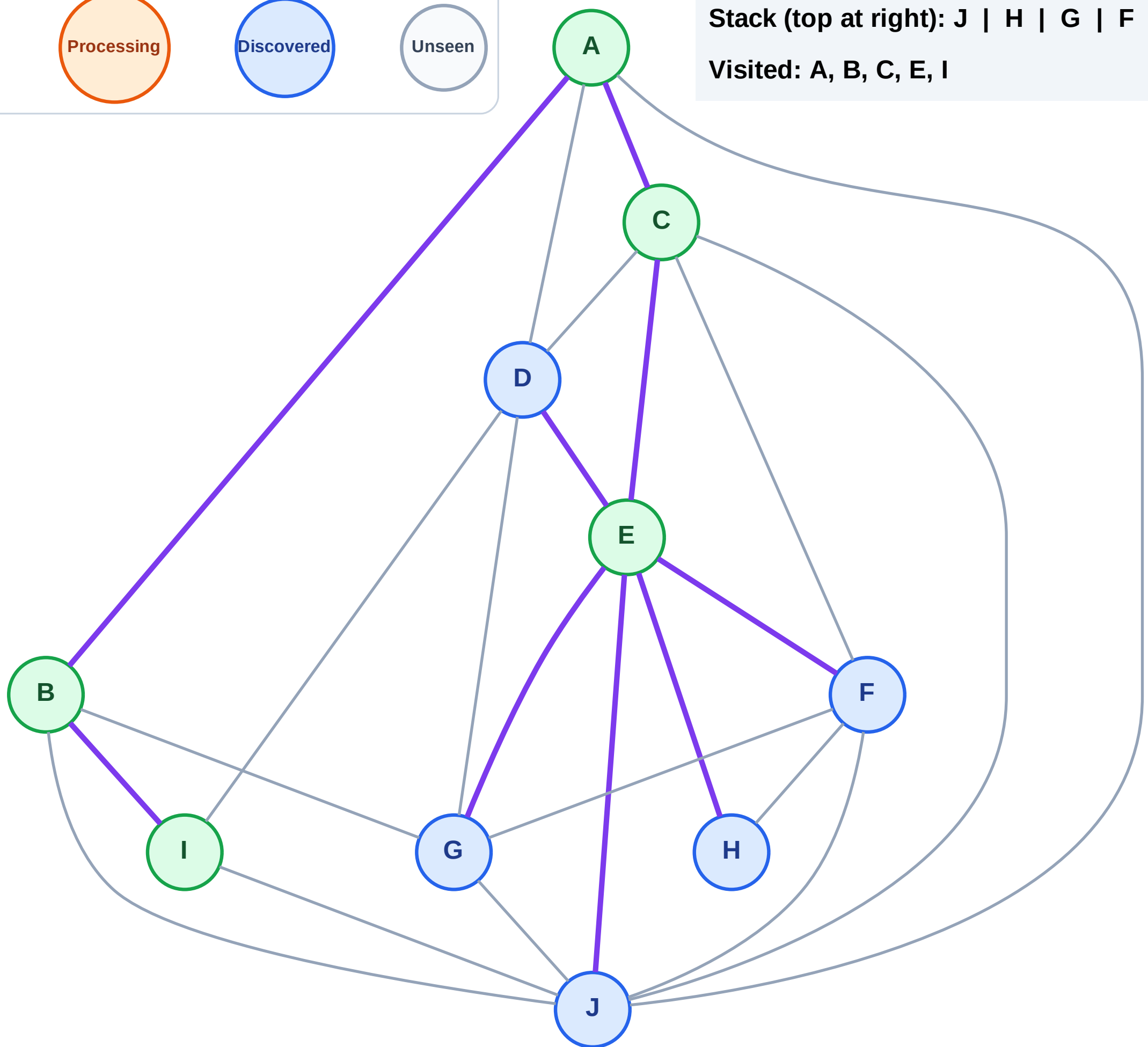
Step 11: No new neighbors from I

## Legend



Stack (top at right): J | H | G | F | D

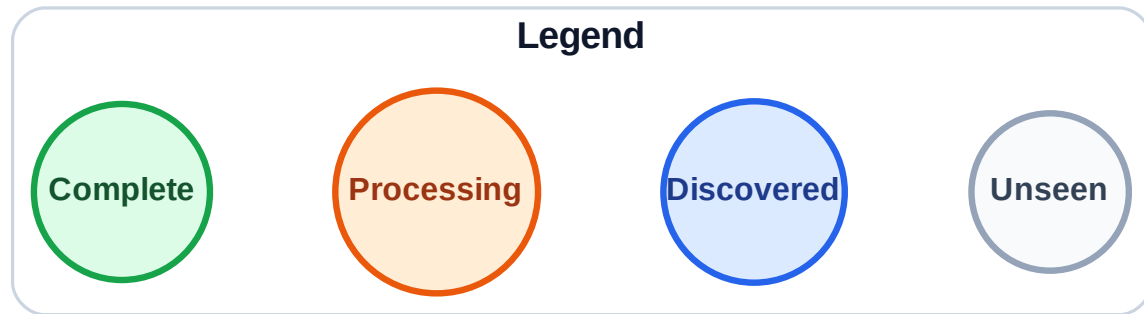
Visited: A, B, C, E, I



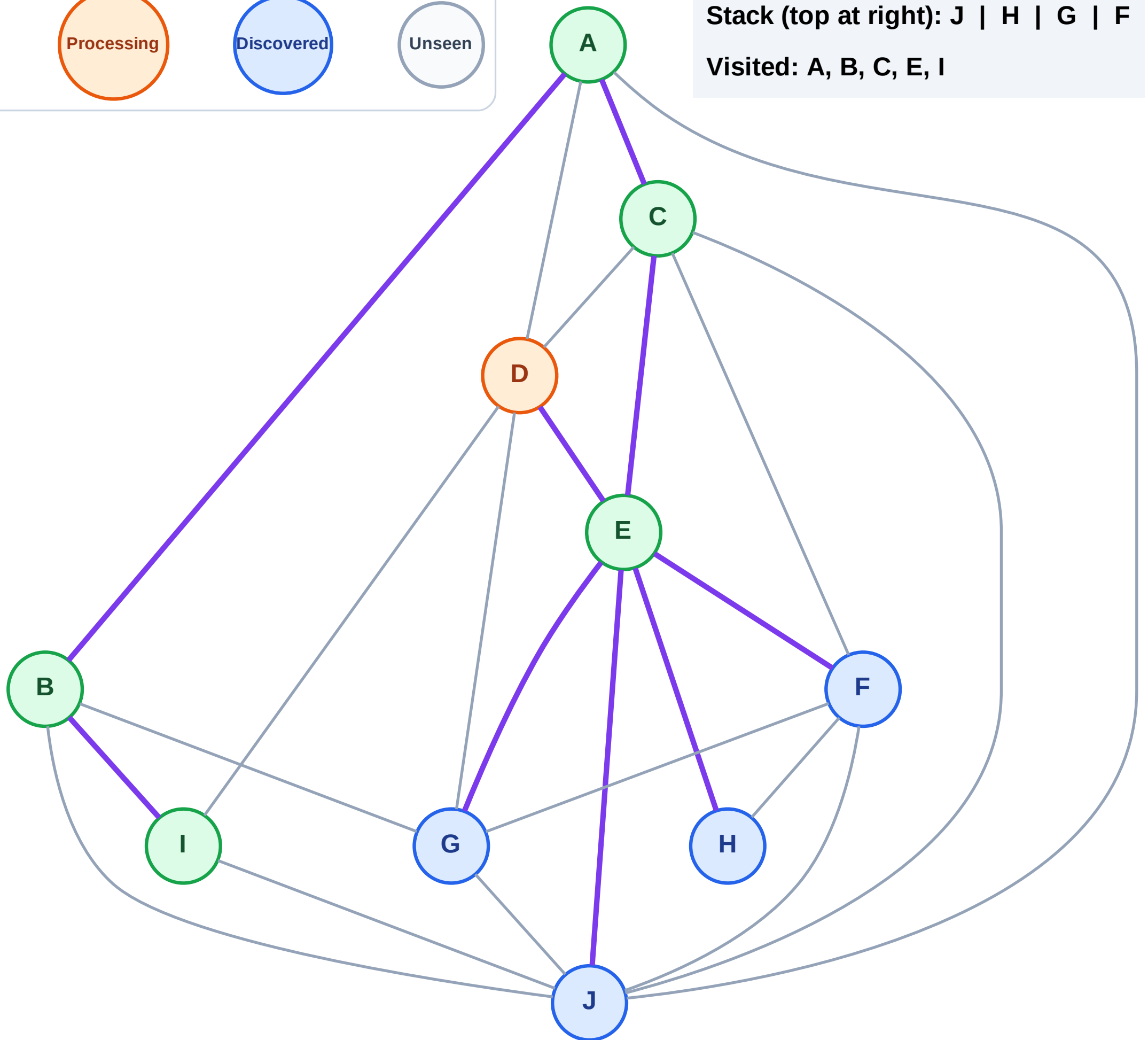


# DFS Traversal

Step 12: Process node D



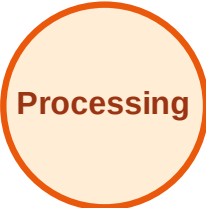
Stack (top at right): J | H | G | F  
Visited: A, B, C, E, I



# DFS Traversal

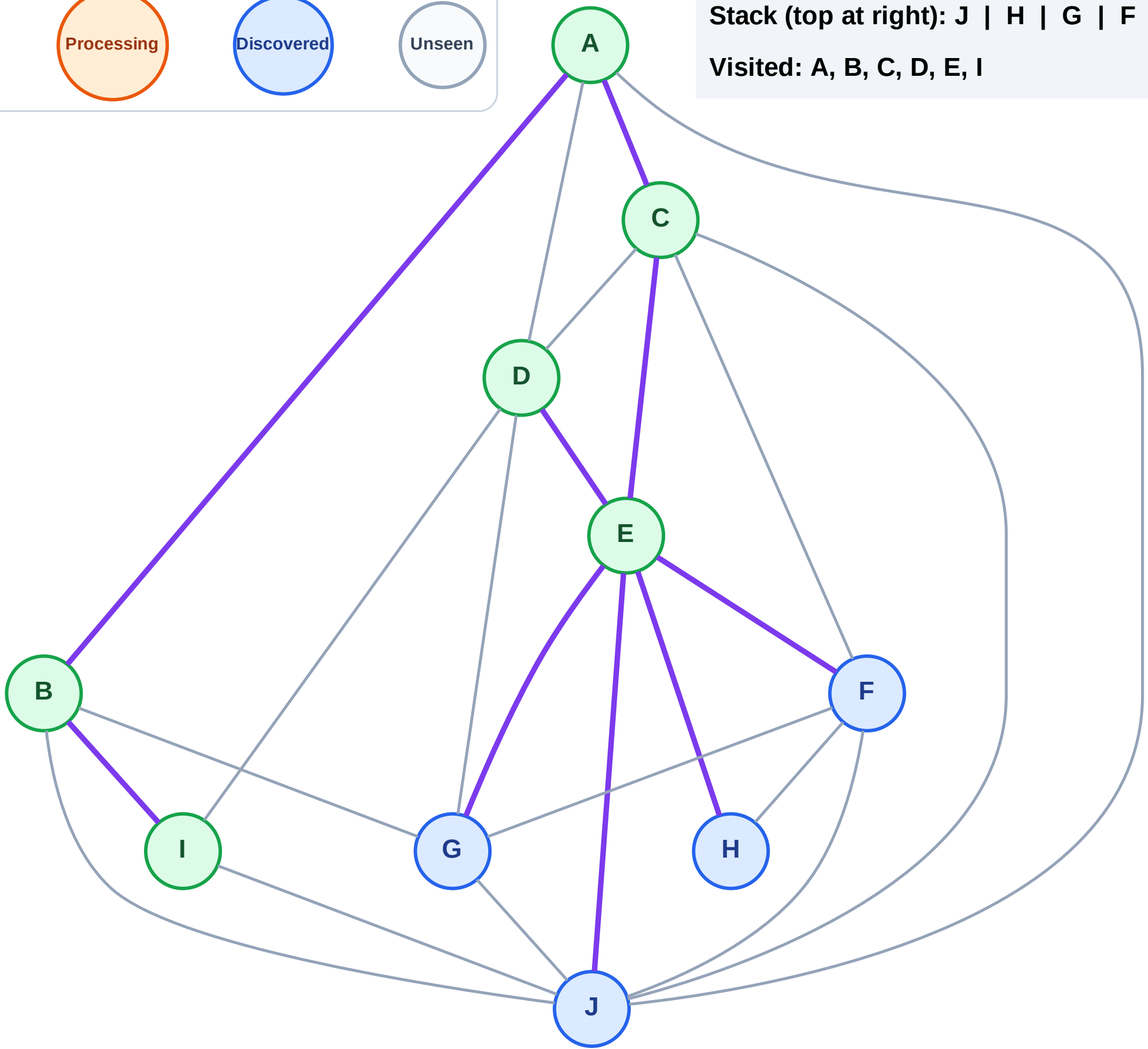
Step 13: No new neighbors from D

## Legend



Stack (top at right): J | H | G | F

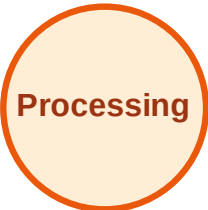
Visited: A, B, C, D, E, I



# DFS Traversal

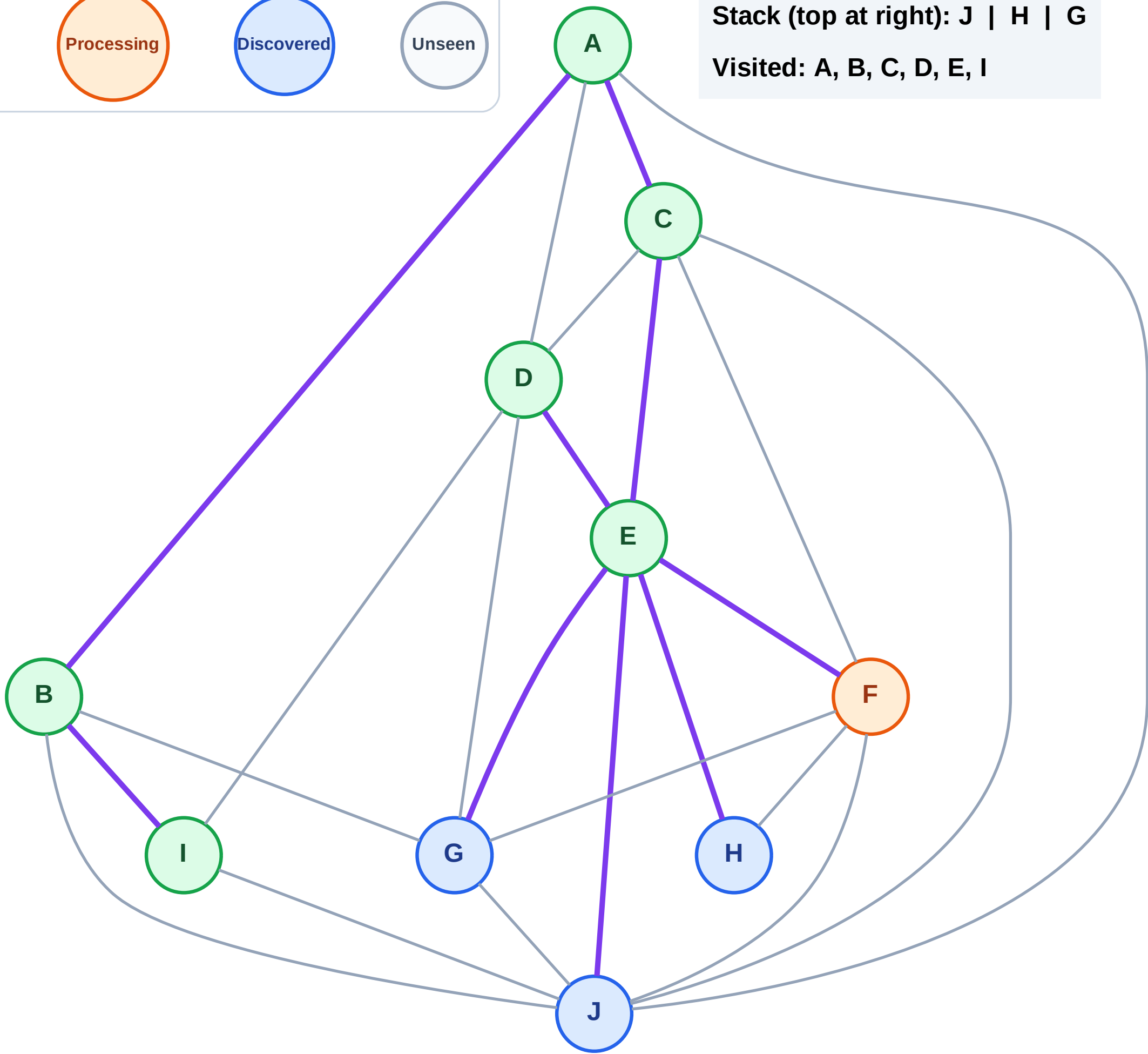
Step 14: Process node F

## Legend



Stack (top at right): J | H | G

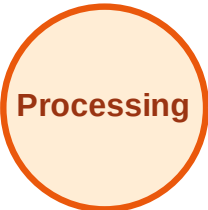
Visited: A, B, C, D, E, I



# DFS Traversal

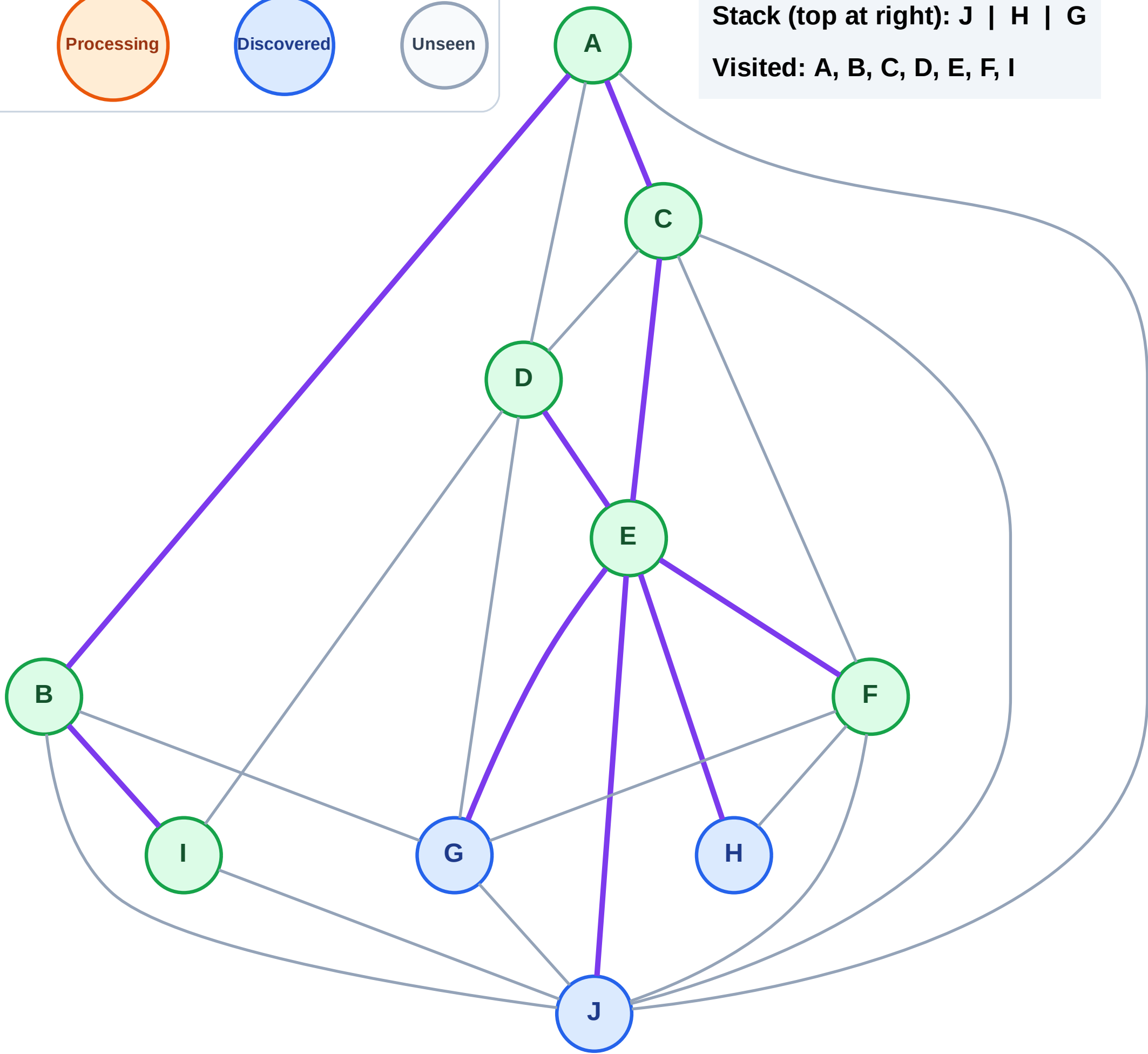
Step 15: No new neighbors from F

## Legend



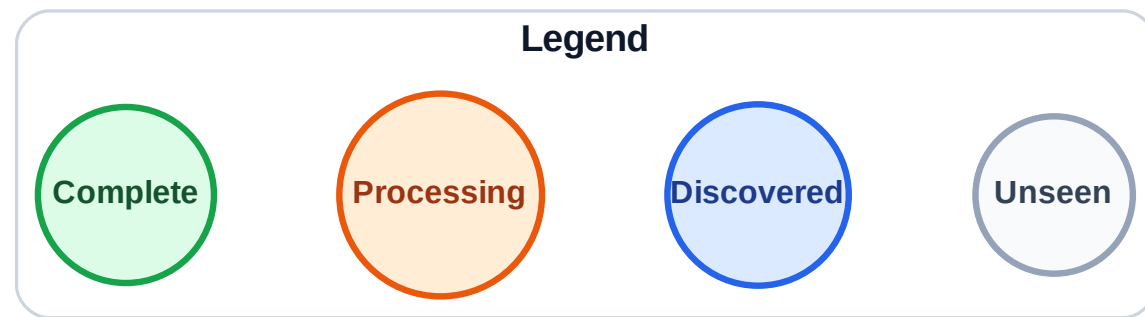
Stack (top at right): J | H | G

Visited: A, B, C, D, E, F, I

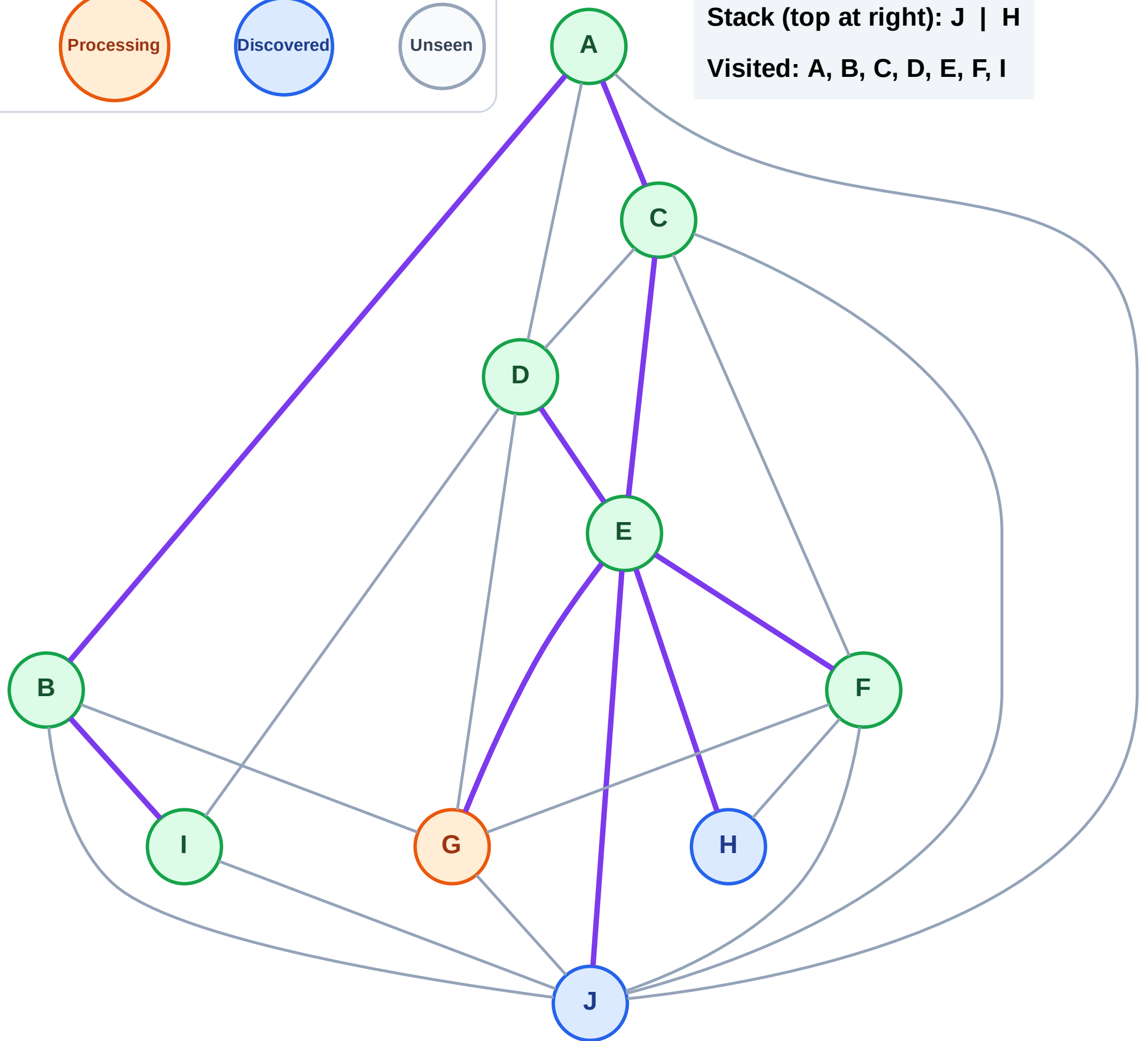


# DFS Traversal

Step 16: Process node G



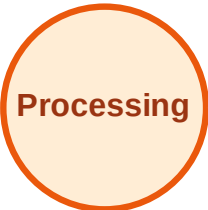
Stack (top at right): J | H  
Visited: A, B, C, D, E, F, I



# DFS Traversal

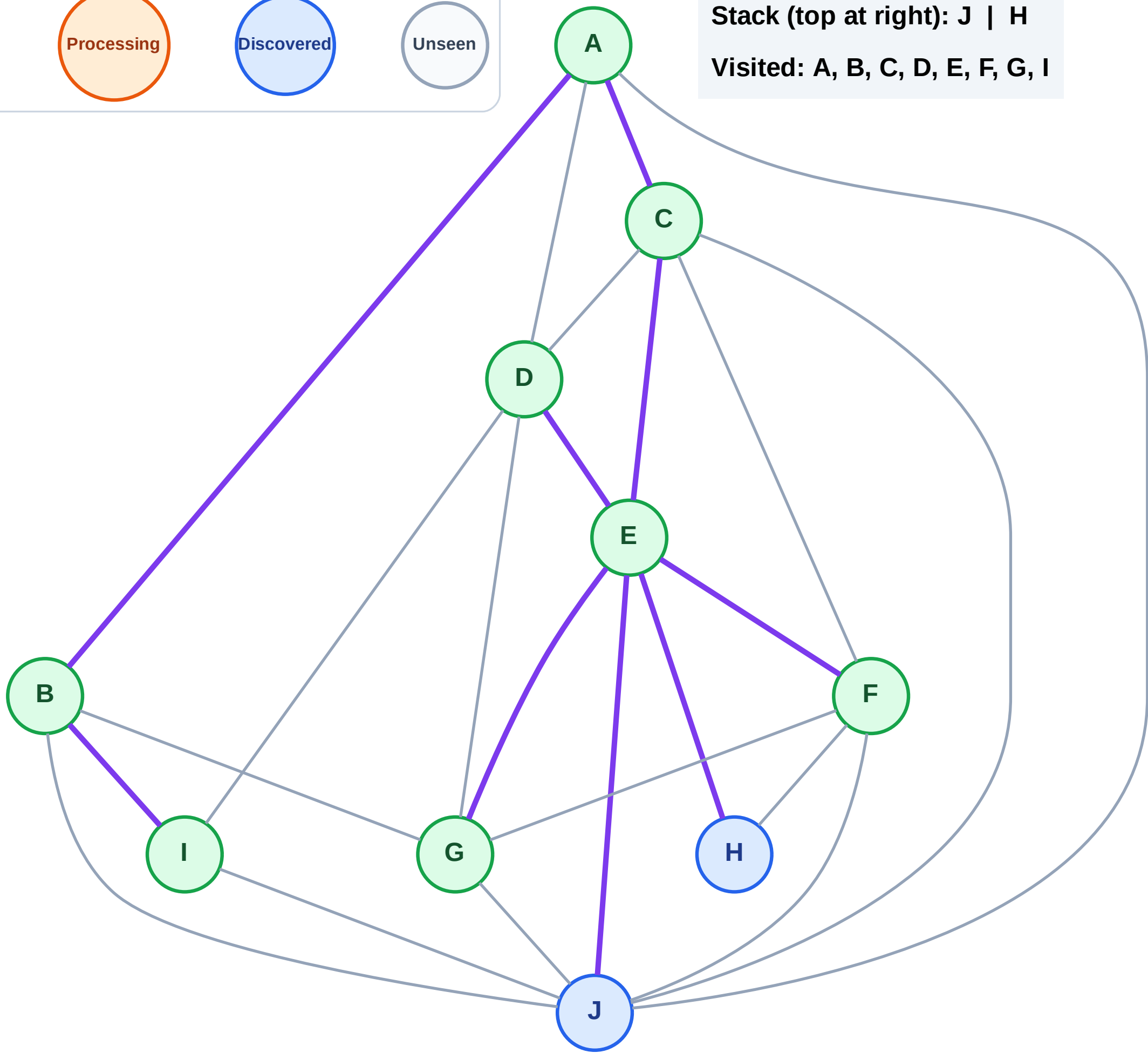
Step 17: No new neighbors from G

## Legend



Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I

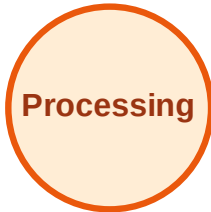




# DFS Traversal

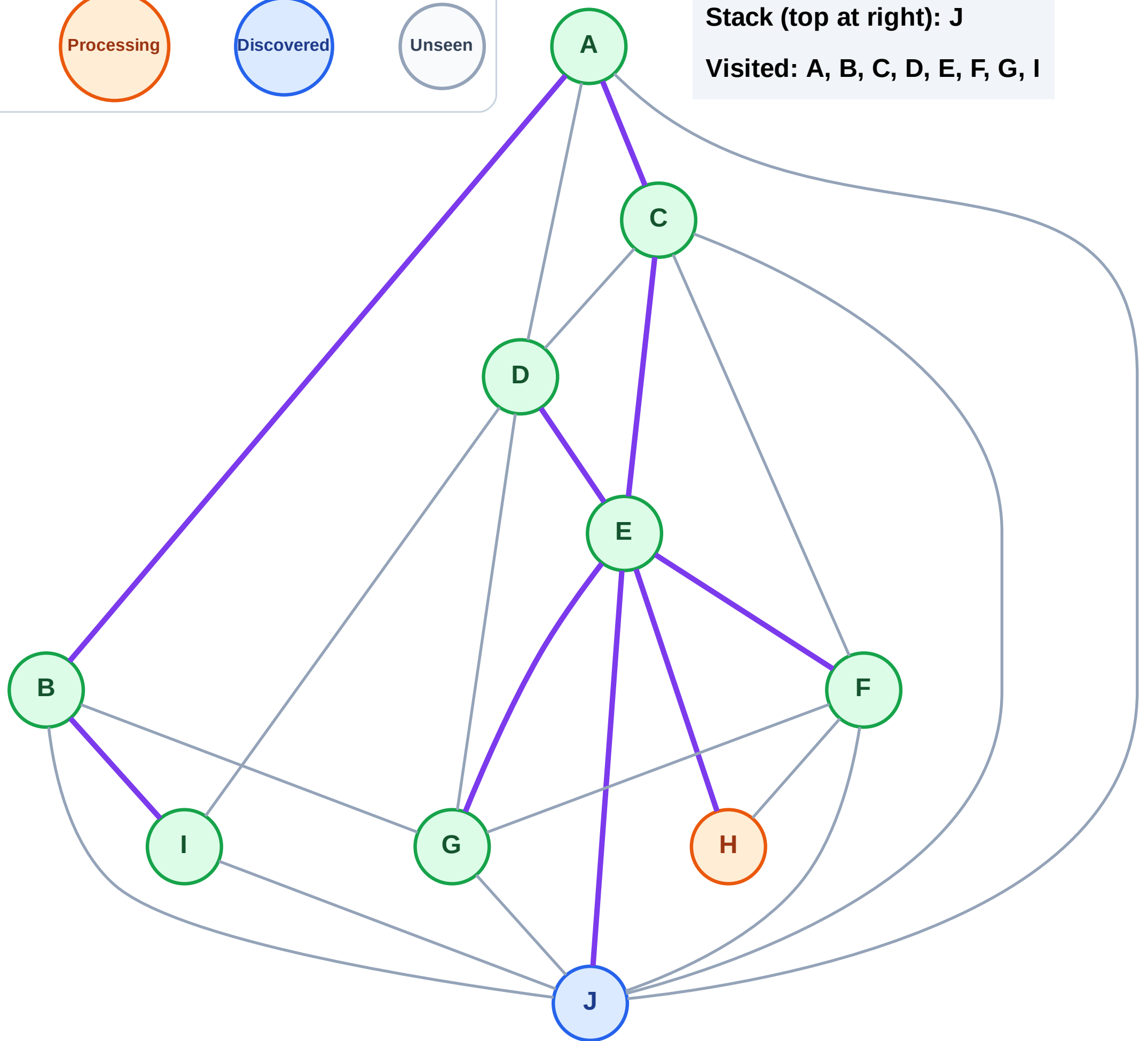
Step 18: Process node H

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, I



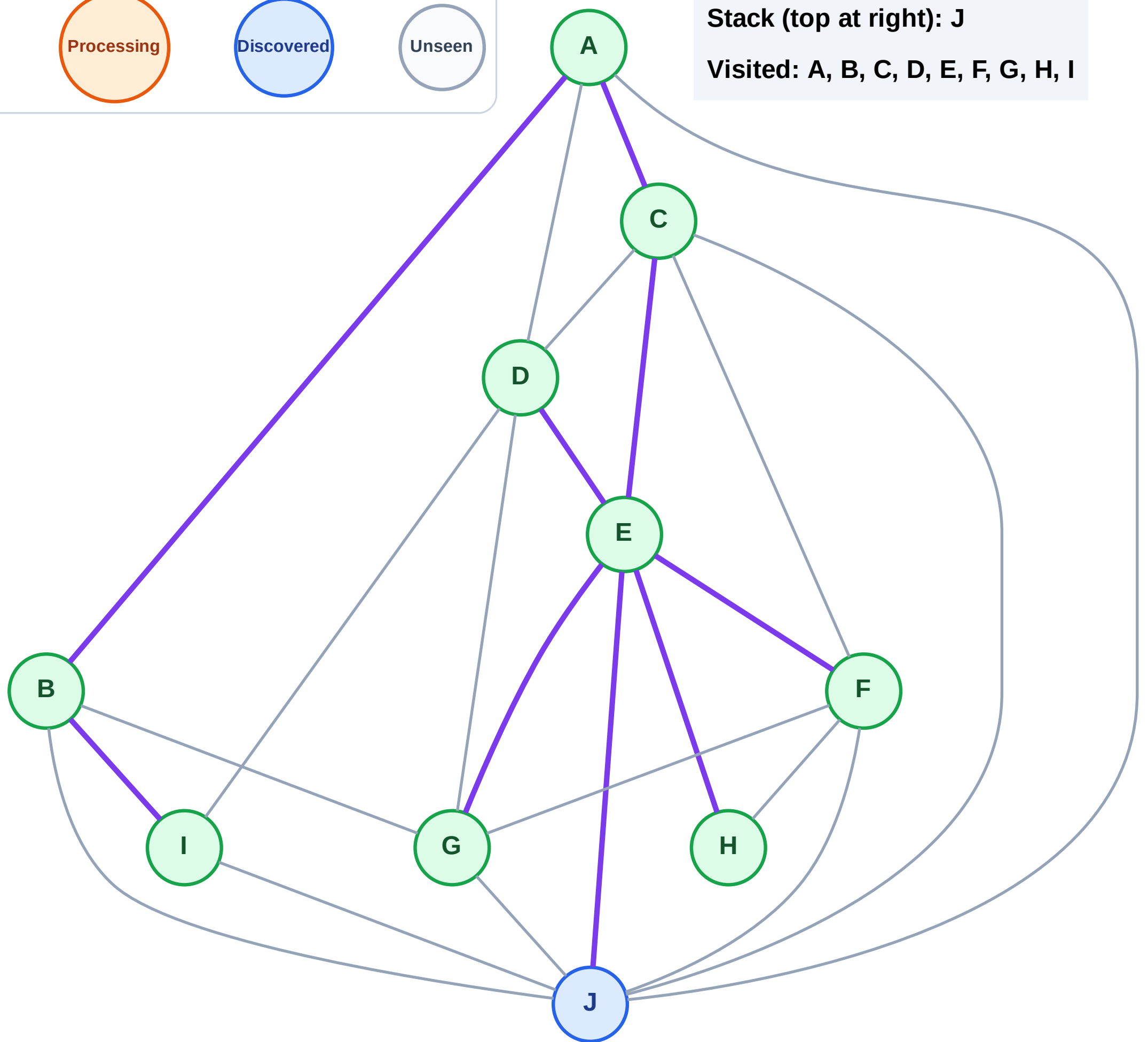
Step 19: No new neighbors from H

**Stack (top at right): J**

### Legend

## Processing

## Unseen





# DFS Traversal

Step 20: Process node J

## Legend



Complete



Processing



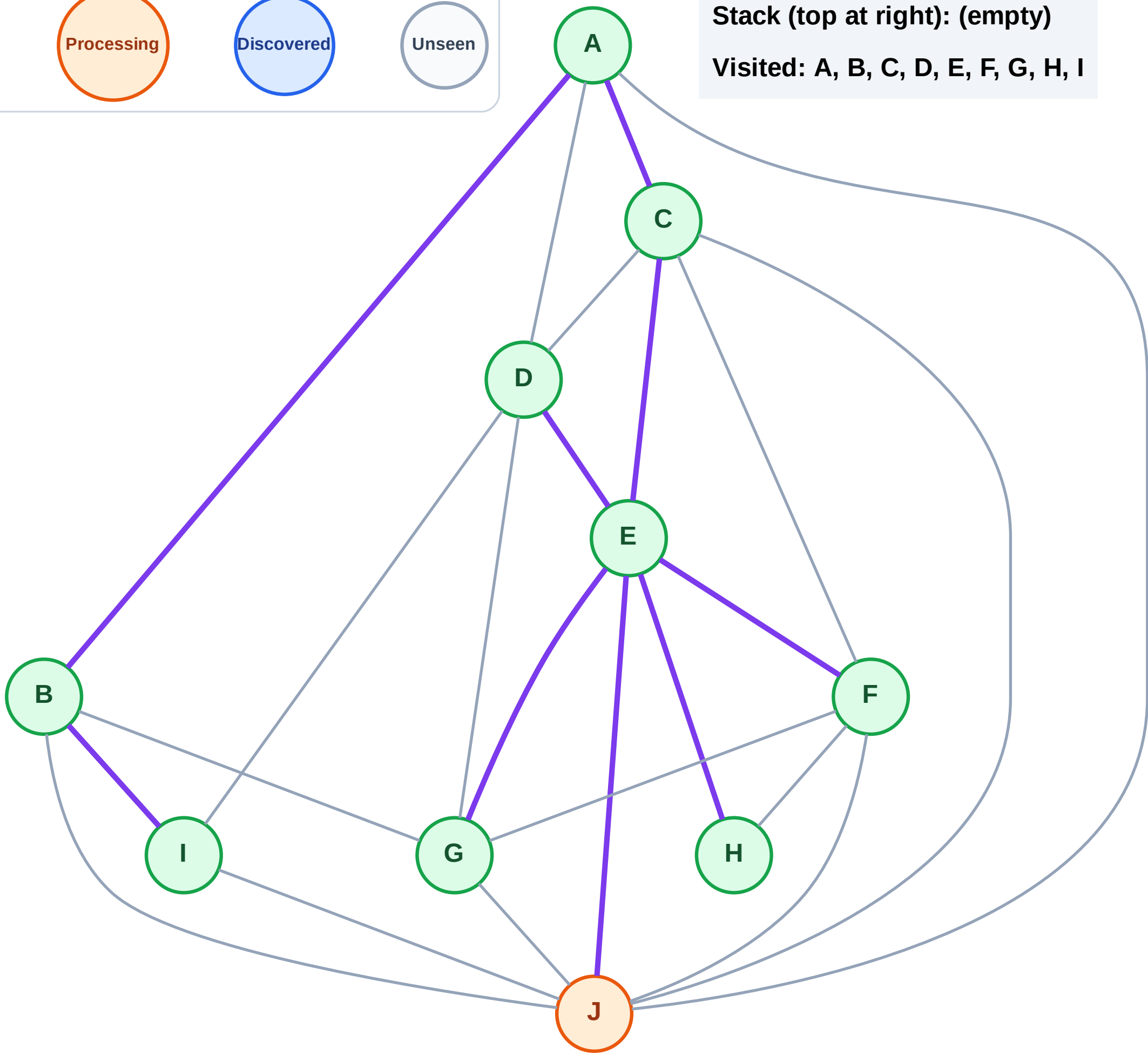
Discovered



Unseen

Stack (top at right): (empty)

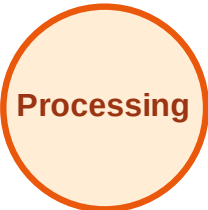
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

